Tutorials on Ray Tracing Simulations for X-ray Optics with SHADOW4

From Sources to Complete Beamlines

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October 22, 2026

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1. Introduction to SHADOW4

1.1. Introduction

SHADOW is a ray-tracing program specially optimized for the design of x-ray optical systems in general and synchrotron radiation beamlines in particular.

SHADOW has demonstrated its reliability during four decades of utilization, supporting beamline design and optimization in most synchrotron facilities from 2nd to 4th generations. It has been used and cited in hundreds of publications.

History SHADOW was created by Franco Cerrina [3] in the early 80's. SHADOW1 (1984-1990), written in Fortran77, was running in VAX-VMS mainframes, and was used not only for the design of monochromators of "Aladdin", a 2nd generation source at the University of Wisconsin, but also at other synchrotrons. SHADOW2 (1990-2010) run in Unix workstation and was also ported to Windows. The ShadowVUI visual interface, available in the XOP environment [11] popularized the use of SHADOW in many facilities and helped in the beamline design at the newly ESRF, a first source of 3rd generation. SHADOW3 [10] was released after a refactoring of the Fortran code to be compatible with new standards of Fortran9x. A Python Application Program Interface (API) [2] was introduced that allowed direct Python scripting and opened the door to new interfaces. Indeed, a quantum leap in user interfacing, that make much easier the beamline design with SHADOW3 was done in ShadowOUI [7]. It put together the power of the SHADOW3 kernel with the beautiful and efficient workflows from ORANGE[1] [6]. Very soon the power of ShadowOUI was boosted with new tools, like the HYBRID method for correcting ray tracing for coherent beams [17], databases for surface error[2], preprocessors for different elements, loops for thermal load calculations [9], and high d-spacing crystals [21]. ShadowOUI was integrated in OASYS [8] a larger suite of programs dedicated to synchrotron applications also including tools for wave optics simulations, the natural complement of ray tracing when dealing with coherent beams.

Just recently a new version, SHADOW4 [5], features a new kernel and a new interface, prepared for new generation of synchrotron sources. The SHADOW4 kernel is a fully object-oriented package named `shadow4`[3], fully implemented in Python. It benefited of the experience and some libraries developed for OASYS. SHADOW4 uses modern and standard software engineering concepts that will permit a larger community of developers to help with new resources, tutorials, and documents. It is naturally interface in OASYS2.

Philosophy SHADOW generates and traces a beam along the optical system. The beam is a collection of rays in a given point of the beamline which are stored in a memory array. The optical system is a collection of optical elements (o.e.) (mirrors, multilayers, slits, screens, etc.) placed in a sequential order.

Each ray is an array of 18 variables or "columns" (they are like a columns in an excel representation of the array of rays, where the row refer to the ray number). Each variable of column has an special physical meaning. The first six defines the geometry: spatial coordinates (Columns 1,2,3 or x, y and z, respectively) and the direction of the ray

[1] `https://orangedatamining.com/`
[2] `https://github.com/oasys-kit/DabamFiles`
[3] `https://github.com/oasys-kit/shadow4`

(cols. 4,5,6, or x',y' and z', respectively). The rest of the columns defines the history of the ray traversing the optical system (electric vector for s-polarization (cols. 7,8,9) and p-polarization (cols. 16-18), flag for lost ray (10), wavelength (11) etc.). In addition to the 18 columns physically stored, there are many other "columns" that are calculated from the native 18. For example, the ray intensity (column 23) is calculated from the electric field components (columns 7,8,9,16,27,18).

Note that the columns 4,5,6 are the direction cosines (the components of the normalized direction vector). In the paraxial approximation, when the beam direction if nearly in the y axis, the direction vector separates from the y axis an angle θ_x (θ_x) along the x (z) axis. In this case, $\theta_x \approx v_x$, $\theta_z \approx v_z$ (see Table).

	Exact	**Paraxial**
a	angle to $\mathbf{e}_x$	$a = 90^\circ - \theta_x$
b	angle to $\mathbf{e}_y$	$b \approx 0^\circ$
c	angle to $\mathbf{e}_z$	$c = 90^\circ - \theta_z$
$\alpha = v_x$	$\cos a$	$\alpha \approx \theta_x$
$\beta = v_y$	$\cos b$	$\beta \approx 1$
$\gamma = v_z$	$\cos c$	$\gamma \approx \theta_z$
identity	$\alpha^2 + \beta^2 + \gamma^2 = 1$	$\theta_x^2 + 1 + \theta_z^2 \approx 1$

Table 1: Direction cosines under the paraxial approximation (propagation close to $\mathbf{e}_y$)

The source is the beam at the starting point. It is generated by SHADOW by sampling the spatial, angular, energy and other qualities of the synchrotron radiations sources (i.e., bending magnets, wigglers and undulators) into a finite number of rays, using a Monte Carlo method. At the source position the intensity of each ray (or better, its probability of observation) is set to 1. This intensity will decrease along the beamline because of the interaction of the ray with the optical elements. The source generated by SHADOW samples linearly the real source, which allows scaling the intensity with the number of photons. There are several kind of sources in SHADOW:

- Grid sources:the rays are not created by Monte Carlo sampling, but just distributed in grids (in cartesian or polar coordinates, and in real or divergence spaces).
- Geometrical sources: the rays are distibuted randomly folowing different distributions either in space (point, rectangular, elliptical, Gaussian, etc.) and in directions of divergence space (collimated, uniform, Gaussian, etc.). Each ray has is monochromatic (it has a single well dephiced photon energy), but polychromatic or white sources can be created sampling the ray energies using the desired energy spectrum.
- Synchrotron sources: Bending Magnet [13], Wiggler and Undulator [15] sources are magnetic structures used for creating synchtrotron radiation in storage rings. SHADOW has very specialised and performant models to simulate these synchrotron sources.

SHADOW traces the source sequentially thought each individual optical element of the optical system. SHADOW solves the intercept of each ray at a given o.e., calculates the output direction and the decrease in intensity. This decrease is calculated for each ray using a physical model (i.e. Fresnel equations for mirrors, Dynamical Theory of the Diffraction for perfect crystals, etc.)

In SHADOW, the calculation of the beam modified by a beamline element can be separated in two parts:

1. the “geometrical" model
2. the “physical" model

The first one deals with the positioning of the rays as it arrives and leaves the element, the intersection with the optical surface, and how the element modifies the directions (e.g. beam reflection by a mirror, or refraction by a lens). The “physical" model deals with the changes of ray's electric fields (that carry out the intensity) due to the interaction with the element.

The operations of the “geometrical" model are grouped in class family `S4OpticalSurface`), whose subclasses contain the traditional surfaces in SHADOW, namely,

- `S4Conic`, for conic surfaces (described by a second-degree polynomial), including planes, spheres, ellipsoids, hyperboloids, paraboloids and the corresponding cylindrical ones [12]
- `S4Toroid`, a toroid (4th-degree equation)
- `S4Mesh`, a generic surface numerically defined over a regular mesh.

The calculation of the reflectivity or transmittivity of the beamline elements requires the computation of material parameters, typically the refraction indices, the photon-material cross sections, the structure factors for crystals, etc. This computation needs to access tabulated data for the materials, that are available in different ways:

1. **Pre-processors.** Traditionally SHADOW works by separating the access to these data from the main kernel code, using the “pre-processor" concept. A pre-processor is a standalone code that will prepare a “material" file for SHADOW. Three pre-processors are available in SHADOW, `PreRefl` containg refraction indices thus used for absorbers, lenses and mirror reflectivity calculations, `Bragg` containing the ingredients of the structure factors thus used by crystals, and `MLayer` containg the materials, refraction indices and structure of a multilatered element, thus used for multilayers. The use of pre-processor file is preserved in SHADOW4. There are good reasons for that: i) compatibility with old versions and re-use of pre-processor files, ii) the SHADOW4 package may work in standalone mode without importing external libraries, and iii) the use of pre-processor file is faster than the direct access to data libraries and is advantageous for loops and scripts. Pre-processors files can be created with the appropriate OASYS widgets.
2. **DABAX** stands for DAtaBAse for X-ray applications. It is a compilation of tables[4] for X-ray applications with a collection of python programs [5] to access and process these tables.
3. **XRAYLIB**[6] [16] is an open-source library that provides a collection of X-ray interaction data and functions for use in scientific computing. It gives researchers and developers easy programmatic access to atomic and nuclear data relevant to

[4] `https://github.com/oasys-kit/DabaxFiles`
[5] `https://github.com/oasys-kit/dabax`
[6] `https://github.com/tschoonj/xraylib`

X-ray physics, including cross-sections, fluorescence yields, line energies, and more. It is fully integrated in SHADOW4 and OASYS2.

The advantages of DABAX is the decoupling of data from calculations — you can easily change the files in the database without needing to change the programs. OASYS widgets include menus that allows the user to choose their preferred data file. The need for fast access and data simplification for X-ray fluorescence applications drove the development of xraylib as a completely new tool — an ANSI C library providing convenient access to a large number of X-ray related databases, with a focus on quantitative X-ray fluorescence applications. The two libraries are complementary: DABAX is a flexible, file-based database system, while xraylib is a more streamlined programmatic library optimized for work that need high speed.

A powerful feature of the OASYS interface to SHADOW4 is the automatic generation of scripts. Every SHADOW4 widget has a “tab" that displays the python script that runs the system. It can be exported and run outside the OASYS environment. It provides to the users the flexibility to customize and tailor calculations to their specifications, as well of performing scans of variables (e.g. for calculating tolerancies) or perform simulations with millions of rays to improve statistics and to compute errors in the parameters.

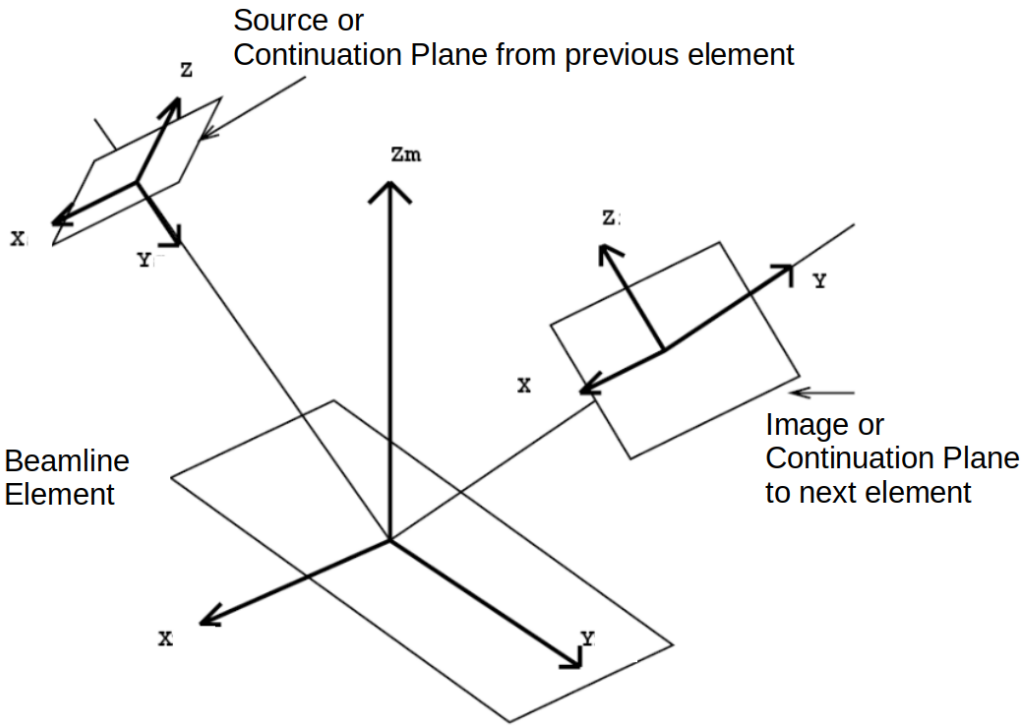


Figure 1: SHADOW4 reference frame definitions: Source, beamline element, and image coordinates.

1.2. Ray tracing examples

This first example is intended to be a first contact OASYS2 and SHADOW4. Just experience using widgets and making simple simulations.

You will:

- define grid and geometrical sources
- Experience with different mirrors
- find the position of foci
- study the image (intensity distribution) at image and on the mirror (footprint)

Questions:

i)

Define grid source over a rectangle in real space and collimated in divergence space

Experience with different mirrors to focus it and find the focal positions. The following mirrors could be studied:

- Plane mirror (no focus)
- Spherical mirror in normal incidence (stigmatic focus)
- Spherical mirror in not-normal (grazing) incidence (astigmatic focus)
- Parabolic mirror, not-normal incidence (perfect stigmatic focus)

ii)

- define geometrical source, point in real space, and isotropic (cone) in divergence space
- place a spherical mirror of R=1 m and distances $p = q = 1$ m. See perfect focus and homogeneous footprint.

Answer

Use the OASYS workspace file `introduction.ows`. Define grid with the center sligtly shifted from (0,0). Define mirrors with internal calculation of the mirror parameters using a focal distance q =0.25 m. Use "FocNew" widget to see the evolution of the beam size, and to check is the focus is at the image position q. Results for Spherical mirror with 45° incidence are at Fig. 1.

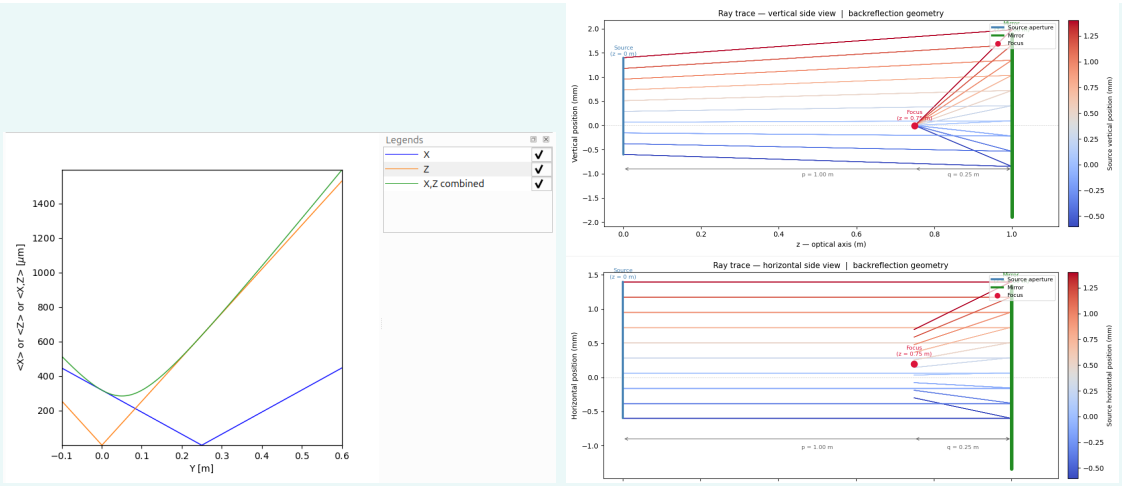


Figure 2: Left: beam evolution (from FocNew). In vertical (Z), the best focus is at zero (at the image plane). In horizontal (X) the best focus is at 0.25 m downstream from the image plane, as a result of astigmatism provoked by the non-normal incidence with spherical reflectors. Right: diagram of the path of the rays.

2. Learning reference frames in SHADOW using a geometrical source

2.1. Reference frame in SHADOW4

SHADOW use local reference systems associated to the source and image planes and to the beamline elements (see Fig. **??**). This permits to iterate over the beamline elements: each one has two "continuation planes": the source and the image.

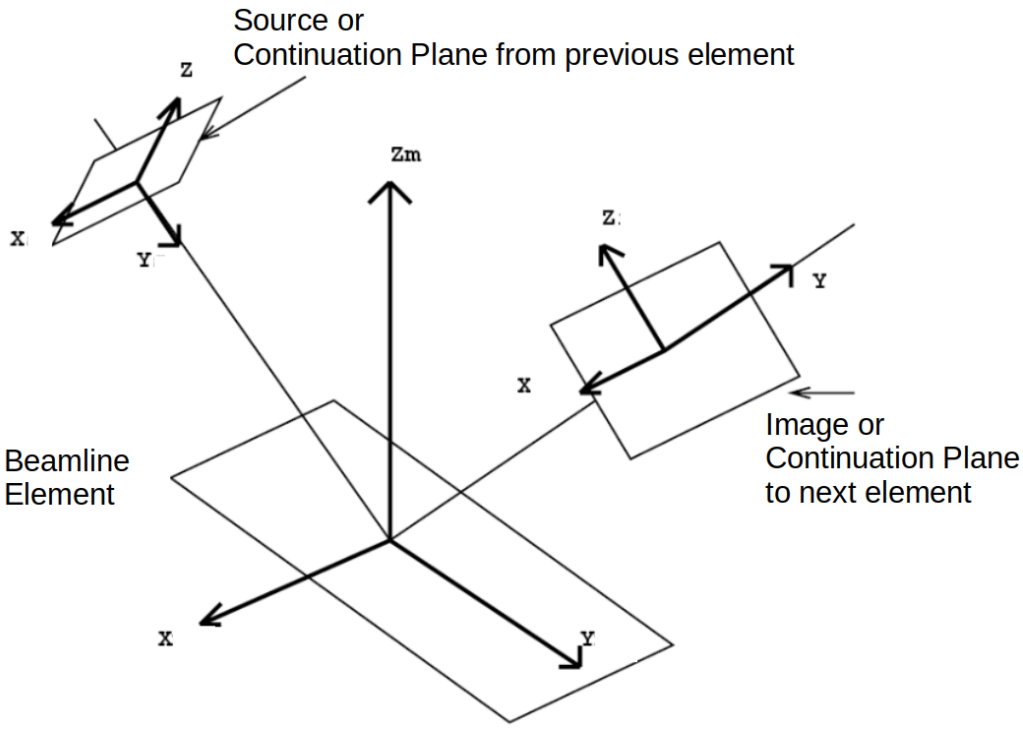


Figure 3: SHADOW4 reference frame definitions: Source, beamline element, and image coordinates.

In SHADOW4 the "beamline element" is formed by an "optical element" object and a "coordinates" object containing two angles: the incidence angle θ and the beamline element orientation angle α, and two distances: the distance p from the source (or continuation plane from the previous element), and q from the beamline element to its image plane or continuation plane. The beamline element orientation angle α is not reset after the scattering in the beamline element (reflection, refraction or diffraction), therefore it influences the orientation of the axis at the image (see Fig. **??**).

2.2. Ray tracing simulations

You will:

- learn to define geometrical sources
- understand the use of a python script for modifying an existing source
- understand reference frames

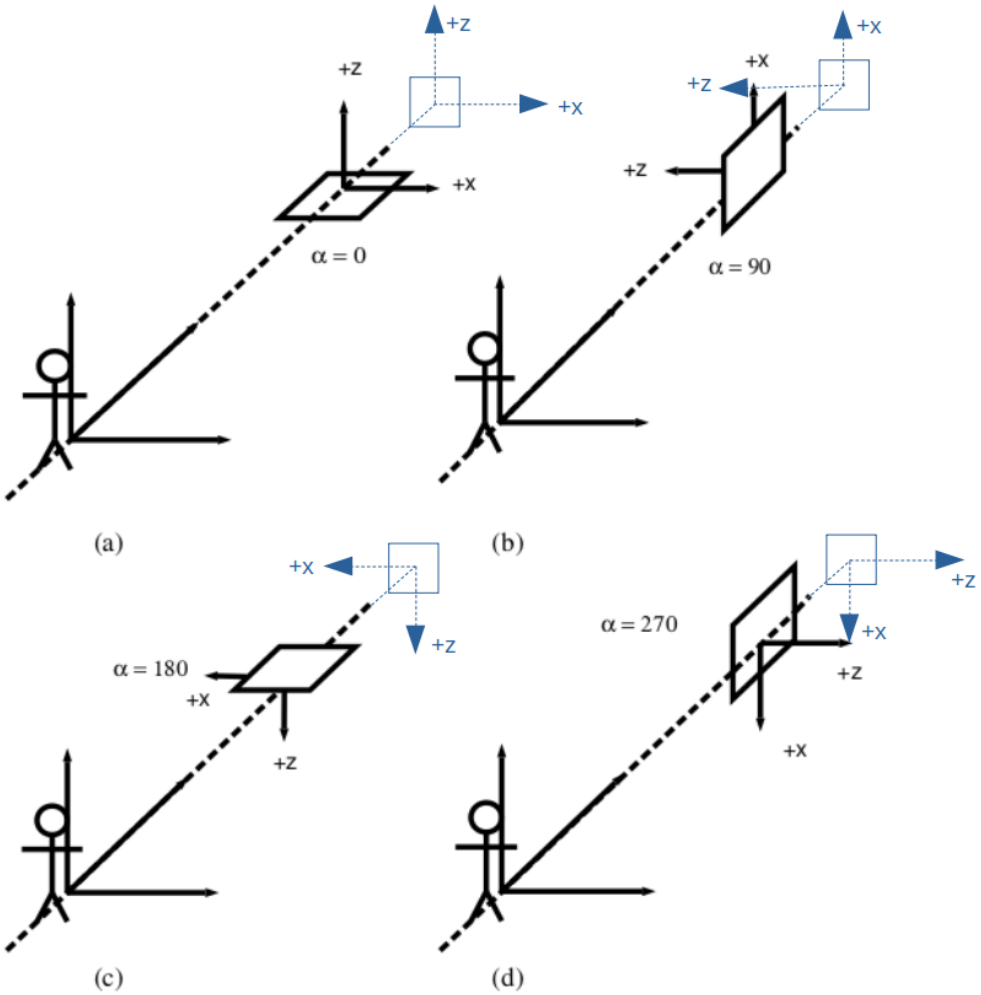


Figure 4: Effect of the beamline element orientation angle α for the four most common values: 0, 90, 180 and 270 degrees. The effect of the angle α is carried out to the image plane, thus defining the position of the axes (in blue).

Questions:

i) Create a collimated (i.e., zero divergence) geometric source with rectangular shape with vertical length 2 mm (X coordinate, column 1) and horizontal length 1 mm (Z coordinate, column 3).

ii) Shift the beam (1, 0, 0.5) mm to get all rays positive.

iii) Create an empty optical element, with incident angle 0 deg (normal incidence) and reflection angle 180° (the same direction), and p=q=0. Trace the system fr optical element orientation angles α equal to 0, 90, 180 and 270 degrees. Verify the results with the pictures shown before.

Hints: you may load the workspace `reference_frame.ows`.

Answer

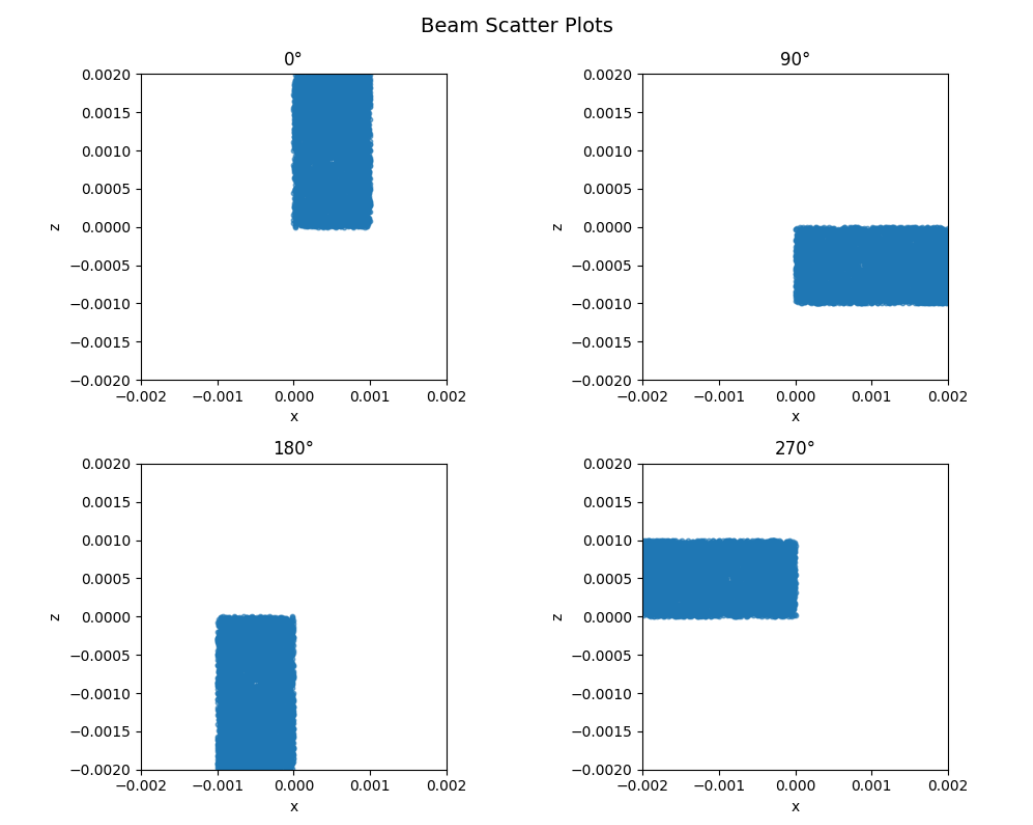


Figure 5: Effect of an empty element in the beam for different α. Note that the beam is unchanged, but the axes are rotated and displayed in the same configuration, therefore there is an apparent rotation of the beam.

3. Synchrotron sources: Bending magnets

3.1. Emission of a Bending-Magnet Synchrotron Source: summary from [4]

A bending magnet (BM) is a dipole magnet that forces relativistic electrons circulating in a storage ring to follow a curved trajectory. The transverse acceleration associated with this circular motion leads to the emission of synchrotron radiation. Bending magnets constitute broadband, incoherent photon sources widely used in synchrotron facilities.

Electron Trajectory and Curvature

An electron of energy E_e moving in a magnetic field B follows a circular trajectory of radius R given by

$$R = \frac{p}{eB} = \frac{m_0 \beta \gamma c}{eB} \approx \frac{3.3356\, E_e\,[\text{GeV}]}{B\,[\text{T}]}, \tag{1}$$

where γ is the relativistic Lorentz factor. Radiation is emitted continuously along the arc of the trajectory.

Spectral Properties

The synchrotron radiation spectrum emitted by a bending magnet is broadband and characterized by a critical photon energy E_c, defined as the energy that divides the total radiated power into two equal halves:

$$E_c = \frac{3\hbar c \gamma^3}{2R} \approx 2.218\, \frac{E_e^3\,[\text{GeV}]}{R\,[\text{m}]}\ \text{keV}. \tag{2}$$

For photon energies $E \ll E_c$, the spectrum rises gradually, while for $E \gg E_c$ it decays exponentially. The spectral shape is described by universal synchrotron functions:

- $G_1(E/E_c)$ for the angle-integrated flux,
- $H_2(E/E_c)$ for the on-axis spectral flux.

Photon Flux

The on-axis spectral photon flux is given by

$$N_0(E) \propto E_e^2\, I\, H_2(E/E_c), \tag{3}$$

while the flux integrated over vertical angles is

$$N(E) \propto E_e\, I\, G_1(E/E_c), \tag{4}$$

where I is the stored electron beam current. Increasing electron energy enhances both the photon flux and the critical energy.

Angular Distribution

Synchrotron radiation from a bending magnet is highly anisotropic. In the vertical plane, emission is confined within a narrow cone of characteristic opening angle

$$\psi \sim \frac{1}{\gamma}. \tag{5}$$

The angular distribution depends strongly on photon energy: low-energy photons are emitted over wider angles, while high-energy photons are more tightly collimated.

In the horizontal plane, photons are emitted tangentially along the curved electron trajectory. In modern low-emittance storage rings, the horizontal extent of the photon source is often dominated by the geometry of the trajectory rather than by the electron beam size itself.

Polarization

Bending-magnet radiation is partially polarized:

- The σ-polarized component (electric field in the orbit plane) is dominant and symmetric about the orbit plane.
- The π-polarized component (electric field perpendicular to the orbit plane) vanishes on-axis and becomes significant off-axis.

The degree of polarization depends on photon energy and emission angle.

Summary

Bending magnets provide broadband, partially polarized synchrotron radiation emitted continuously along a curved electron trajectory. The emission exhibits strong spectral and angular dependencies governed by the critical energy. In modern low-emittance storage rings, geometrical effects play a crucial role in determining the effective photon source size, particularly in the horizontal direction.

3.2. Ray tracing simulation

You will:

- simulate bending magnets

Questions:

i) Simulate the source for the ESRF1 bending magnet (the old storage ring, not longer existing) at a fixed energy (e.g.,. 8 keV). Use one mrad of horizontal divergence. Visualize the cross section (x,z), the divergence space (x',z'), the phase spaces (x,x') and (z,z'). Visualize the top view (y,x). Make histograms of intensity (total, s-polarized and p-polarized) as a function of the vertical divergence. Plot also the degree of circular polarization (S3 component of the Stokes vector).

ii) Change the source energy to 18 keV and compare the plot of intensity versus vertical divergence with the result at 8 keV. Verify that the radiation is more collimated.

Hints: you may load the workspace `bendingmagnet.ows`.

Answer

Use the "Bending Magnet" widget, click "Run shadow4/source" and display in the right panel using "Detailed Plot" the cross section (x,z), the divergence space (x',z'), and phase spaces (x,x'), (z,z').

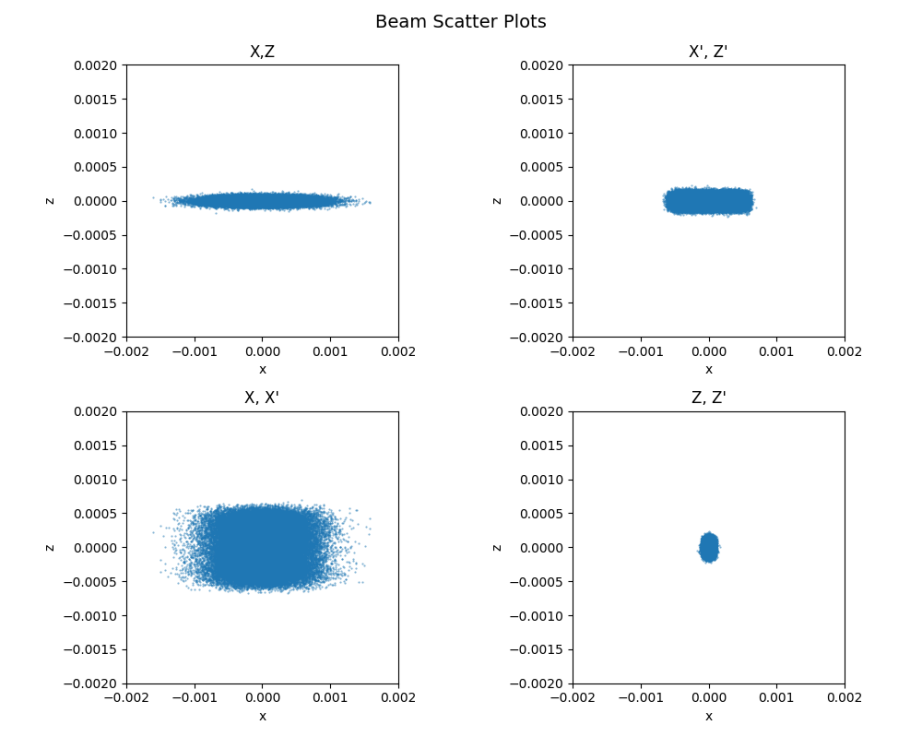


Figure 6: Rays in real space (X,Z), divergence space (X',Z'), phase space (X,X') along column 1, and phase space (Z,Z') along column 2.

Another way is to plug a "Plot XY" widget and select the wanted coordinates for the plot. This can also be used to display the "top view" (y,x).

Plug the "Histogram" widget to display histograms of intensity weighted by 23:Total Intensity, 24:s-polarized, 25: p-polarized, and 33: S3-Stokes or circular polarization. Although it is not possible to combine histograms in a single plot, it is always possible to prepare a simple script that performs customized plots, like the one included in the workspace that produces the outpur in figure 2.

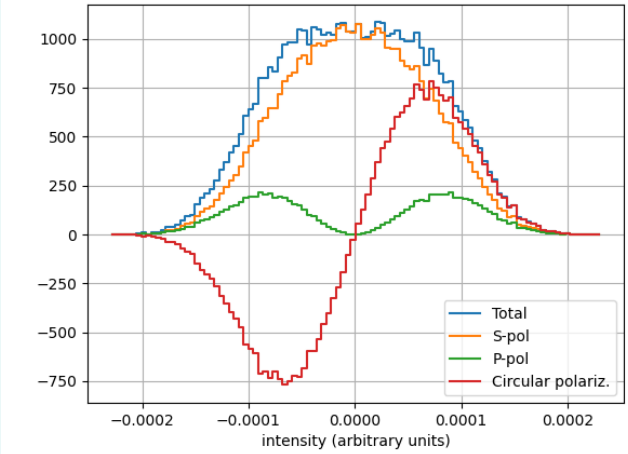


Figure 7: Histograms of the total intensity (weight column 23), S-polarization (weight column 24), P-polarization (weight column 25) and S3-Stokes (weight column 33), versus Z'.

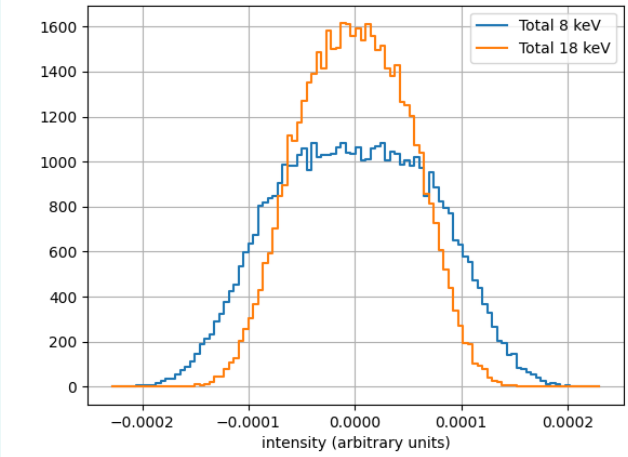


Figure 8: Histograms of the total intensity vs Z' for emission at 8 and 18 keV.

4. Synchrotron sources: Wigglers

4.1. Synchrotron Wiggler Emission: a summary

Wigglers and short insertion devices (IDs) are increasingly used as photon sources in modern synchrotron facilities, particularly in upgraded storage rings like ESRF-EBS. Unlike undulators, wiggler radiation is **fully incoherent**, making existing undulator calculation methods inefficient. The model used in SHADOW4 is summarized here.

Key Assumptions

The method assumes:

1. **Fully incoherent emission**: Radiation from different points along the electron trajectory superimposes incoherently.
2. **Bending magnet approximation**: Emission at each trajectory point equals that of a bending magnet with the same local curvature.

Calculation Method

Electron Trajectory Calculation

The wiggler magnetic field for a conventional wiggler has only a vertical component $(0, 0, B_z)$. Suppose we know a tabulation of $B_z(y)$ (measured or calculated, e.f. sinusoilal). An electron entering in this magnetic field will have a transversal velocity given by the integral of the magnetic field

$$\beta_x(y) = -\frac{c \times 10^{-9}}{E_e[GeV]} \int_{y_0}^{y} B_z(s)\, ds. \tag{6}$$

The electron trajectory is in the horizontal plane and is given by the integral of the velocity

$$x(y) = \int_{y_0}^{y} \beta_x(s)\, ds. \tag{7}$$

Note that in both integrals an additional constant can be added in order to obtain "centered" profiles. This is discussed in the practical exercise, and it will be whown that an adequate selection of the constant is important to get the radiation centered on axis.

Local Curvature and Critical Energy

The local curvature (or inverse of the local radius) is also calculated, because the local emission corresponds to that of a bending magnet of that curvature. It is proportional to the second derivative of the trajectory, therefore proportional to B_z

$$R^{-1}(y) = -\frac{|e|}{\gamma m_e c} B_z(y) \tag{8}$$

with e the electron charge, m_e the electron mass and γ the electron energy in units of $m_e c^2$. Each curvature has associated a bending magnet critical energy given by

$$E_c(y) = \frac{3}{4\pi}\lambda\gamma^3 R^{-1}(y) \tag{9}$$

Wiggler Spectrum Calculation

The total spectrum $F(E)$ is obtained by:

1. Calculating bending magnet emission at each z point
2. Multiplying by $(R(z)10^{-3})^{-1}$ for arc length conversion
3. Accounting for trajectory projection via $\sqrt{1+(\beta_x(z)/\beta_z(z))^2}$
4. Integrating over z

Sampling mechanism

The rays are obtained by:

1. Sample the energy
2. Sample the position in the trajectory
3. Add electron emittance

Conclusion

The presented algorithm provides a fast, accurate method for ray tracing simulations of wigglers and short insertion devices. It efficiently handles the incoherent nature of wiggler radiation and accommodates non-sinusoidal magnetic fields.

4.2. Ray tracing simulations

You will:

- Simulate wigglers

Questions:

i) Simulate the wiggler for the ESRF ID19 in the energy interval 10000±10 eV. Estimate the total horizontal divergence looking at the velocity plot and compare with the width of the x' histogram. Visualize a top view of the emission (y,x) with and without emittance. Input data is in Table 2.

Table 2: Electron and wiggler Parameters for ID19.

Parameter	Value	Unit
σ_x	31.16	μm
σ_y	5.16	μm
$\sigma_{x'}$	4.52	μrad
$\sigma_{y'}$	1.94	μrad
Electron beam energy	6.0	GeV
Electron current	0.200	A
Period length	0.15	m
Number of periods	6.867	—
Horizontal K	0.0	—
Vertical K	20.4911	—

Hints: you may use the workspace file `wigglers.ows`

Answer

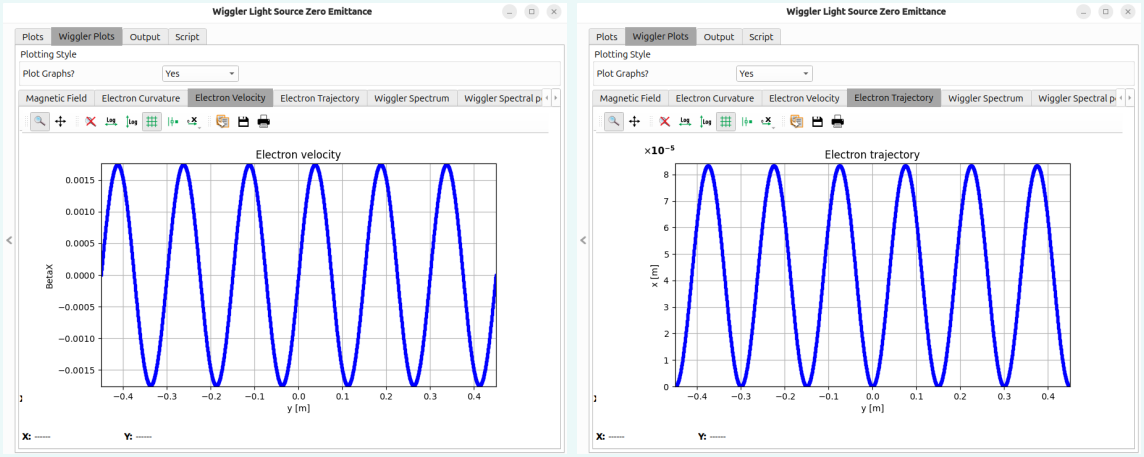


Figure 9: Electron transversal velocity (left) trajectory (right). Note that the velocities are centred, but not the trajectories. By default, the electron enters the wiggler at $x = 0$.

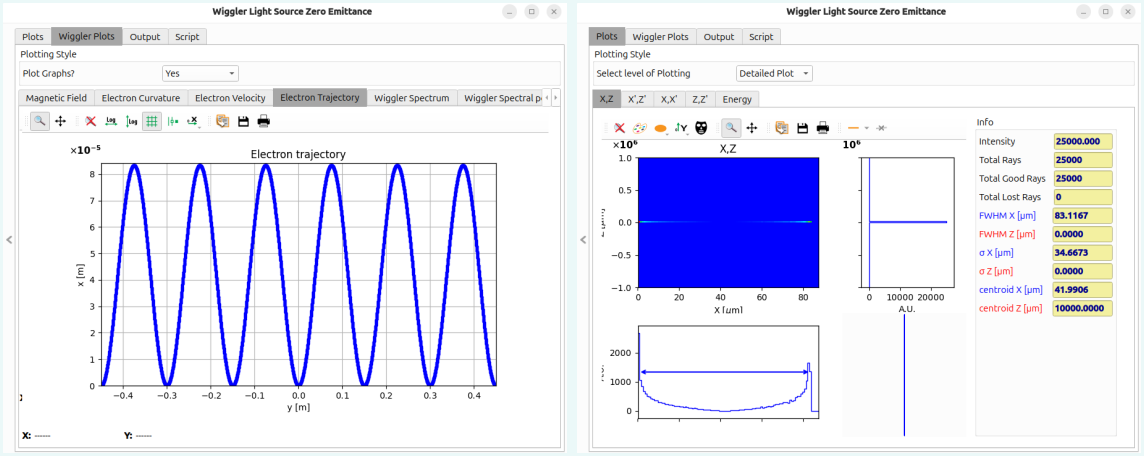


Figure 10: Electron trajectory (left), and source emission. Note that the non-centered trajectory imply that the photon source is not centered at zero. This is “corected" in next figure.

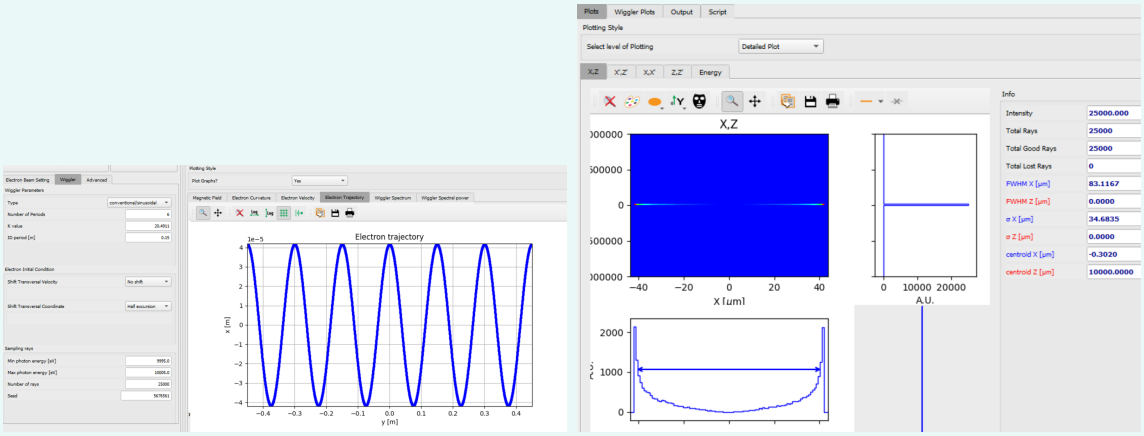


Figure 11: Electron trajectory corrected to place the center at “half-excursion" (left), and source emission. Note that the centered trajectory imply that the photon source is now centered at zero.

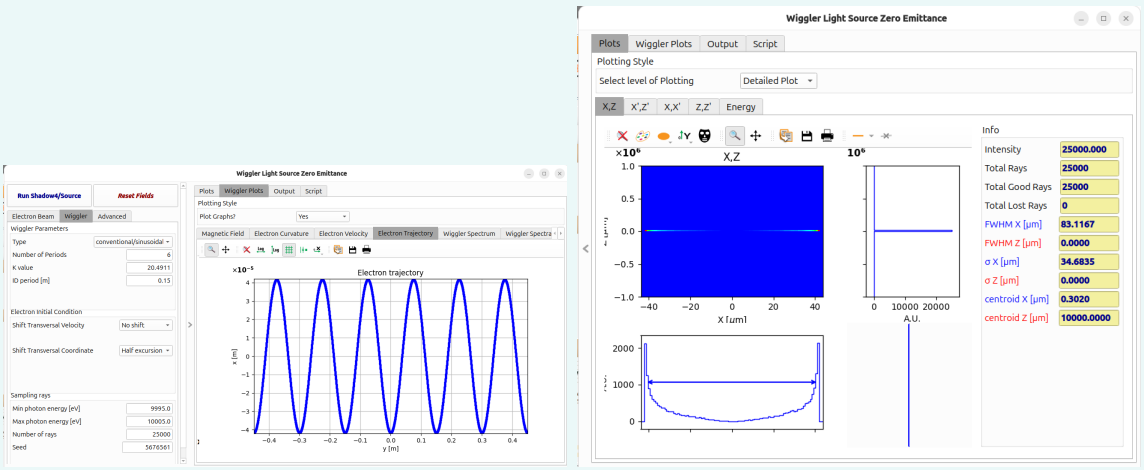


Figure 12: Phase space (X,X’) at the source. Left: The default play displays the (X’, Z’) coordenates, therefore ignoring the effect Y coordinate (depth). Right: by doing "retrace" the rays are propagated to the plane $y = 0$, thus considering the depth of the source. This produces the typical “star" phase-space in the wigglers.

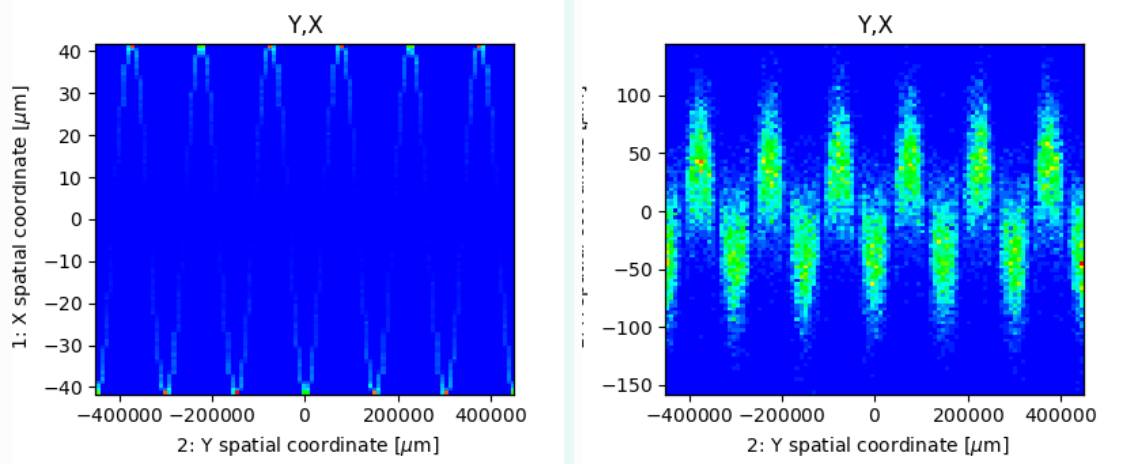


Figure 13: Top view of the emission $x(y)$ for the wiggler placed in a zero emittance ring (left), or in a ring with emittance (right)

5. Synchrotron sources: Undulators

5.1. Synchrotron Undulator Emission: a summary based on [14]

Basic Principles

Undulators produce brilliant, collimated synchrotron radiation through periodic electron trajectories in a sinusoidal magnetic field.

$$K \approx 93.37 \cdot B[T] \cdot \lambda_u[m] \tag{10}$$

where K is the deflection parameter, B is the magnetic field, and λ_u is the undulator period.

The resonance wavelength (on-axis, $\theta = 0$) is:

$$\lambda = \frac{1 + K^2/2}{2\gamma^2}\lambda_u \tag{11}$$

where $\gamma \approx 1957 \cdot E[GeV]$ is the Lorentz factor.

Angular Distribution (Divergence)

At resonance, the far-field intensity follows a sinc^2 pattern:

$$I(\theta) \propto \text{sinc}^2\left(\frac{\pi}{2}\tilde{\theta}^2\right), \quad \tilde{\theta} = \theta\sqrt{L/\lambda} \tag{12}$$

where $L = N\lambda_u$ is the undulator length.

The Gaussian approximation for divergence (r.m.s.) is:

$$\sigma_{r'} \approx 0.69\sqrt{\frac{\lambda}{L}} \quad \text{(Elleaume model)} \tag{13}$$

Source Size (Spatial Distribution)

Obtained by backpropagating far-field radiation to the source plane ($z = 0$):

$$\sigma_r \approx 0.218\sqrt{\lambda L} \tag{14}$$

The phase space product (non-Gaussian) is:

$$\sigma_r \sigma_{r'} \approx \frac{\lambda}{2\pi} \quad \text{(approximately twice the diffraction limit)} \tag{15}$$

Electron Beam Effects

The electron beam is described by a 5D Gaussian distribution:

$$(x, x', y, y', \delta_\mathcal{E}) \tag{16}$$

with sizes σ_x, σ_y, divergences $\sigma_{x'}, \sigma_{y'}$, and energy spread $\delta_\mathcal{E}$.

The photon source size and divergence including electron effects:

$$\Sigma_x = \sqrt{\sigma_r^2 + \sigma_x^2}, \quad \Sigma_{x'} = \sqrt{\sigma_{r'}^2 + \sigma_{x'}^2} \tag{17}$$

Energy spread correction (Tanaka model):

$$\Sigma_{x'} = \sqrt{(Q_a \sigma_{r'})^2 + \sigma_{x'}^2}, \quad \Sigma_x = \sqrt{(Q_s \sigma_r)^2 + \sigma_x^2} \tag{18}$$

where Q_a and Q_s are corrective functions depending on:

$$\sigma_e = 2\pi n N \delta_{\mathcal{E}} \tag{19}$$

with n = harmonic number and N = number of periods.

SHADOW4 Undulator Models

A. "Undulator Gaussian" (Simplified)

- Assumes monochromatic source at/near resonance
- Rays sampled from Gaussian distributions
- Includes optional energy spread correction
- Useful for quick beamline prototyping

B. "Undulator Light Source" (Full Model)

1. Compute far-field electric field stack (3D: energy, θ, φ) for filament beam
2. Apply energy spread correction to angular array (if monochromatic)
3. Use intensity stack as PDF to sample ray energies and directions
4. Add electron emittance via Gaussian convolution
5. Backpropagate far field to source plane for spatial distribution
6. Sample ray positions from convolved distribution
7. Assign polarization from interpolated far-field data

Important for 4th-Generation Sources

- **Diffraction-limited size** becomes significant with low electron emittance
- **Energy spread effects** are crucial for high harmonics
- **Non-Gaussian phase space** requires careful modeling beyond Gaussian approximations
- **Backpropagation methods** are essential for accurate source size calculation

Conclusion

Undulator radiation is highly collimated and peaked at harmonic energies. Accurate ray-tracing requires proper treatment of electron beam parameters, non-Gaussian phase space modeling, backpropagation for source size calculation, and energy spread corrections for high harmonics. SHADOW4 implements both simplified Gaussian and full wave-optics-based models for modern synchrotron beamline design.

5.2. Ray tracing simulations

You will:

- Simulate undulators using different approximations.

Questions:

i) Simulate the ESRF ID02 U21 undulator with gap tunned to have its first harmonic at 15 keV. Then run the simulation at 9th harmonic, or energy $E_9 = 9E_1 = 135$ keV. Use Gaussian and neglect electron energy spread.

ii) Activate the use of the electron energy spread, set $\Delta E_e/E_e = 0.001$. Re-calculate 1st and 9th harmonics at (15 and 135 keV; do not forget to set now set n=1 and n=9 in the "Advanced Settings"). Explain the differences.

iii) Using the "Undulator Light Source" widget (not the Gaussian approximations), calculate the monochromatic emission at three energies: $E_1 \pm 10$ eV. Explore methods of calculating source size (via back-propagation).

iv) Using the "Undulator Light Source" widget (not the Gaussian approximations), calculate the polychromatic emission at $E_1 \pm 10$ eV.

Table 3: Electron Parameters (retrieved from S28F for ID02)

Parameter	**Value**	**Unit**
ID02 U21-4b		
σ_x	33.39	μm
σ_y	7.28	μm
$\sigma_{x'}$	4.53	μrad
$\sigma_{y'}$	1.94	μrad
Electron beam energy E_e	6.0	GeV
Electron current	0.200	A
Period length	0.0214	m
Number of periods	74.766	—
e$^-$ energy spread $\Delta E_e/E_e$	0.001	
Number of periods	74.766	—
Horizontal K	0.0	—
Vertical K	0.360584	—
Length L	1.6	m
Resonance (1st harm) E_1	15	keV
Resonance (9th harm) E_9	135	keV

Hints: you may use the workspace file `undulator.ows`

Answer

i) Use the "Undulator Gaussian" widget and input the corresponding parameters for the electron beam and photon energy. This application will create a source using the

Shadow Geometrical source with divergences and sizes corresponding to the photon beam (eqs. 14 and 13). These values or the photon beam comes from a convolution (sum in quadrature) of the values for the electron beam with the values corresponding to the photon emission of a single electron beam as shown in equations 17 or 18.

The plots of the histograms for the real (X,Z) and divergence spaces (X',Z') for the undulator are in figure 14.

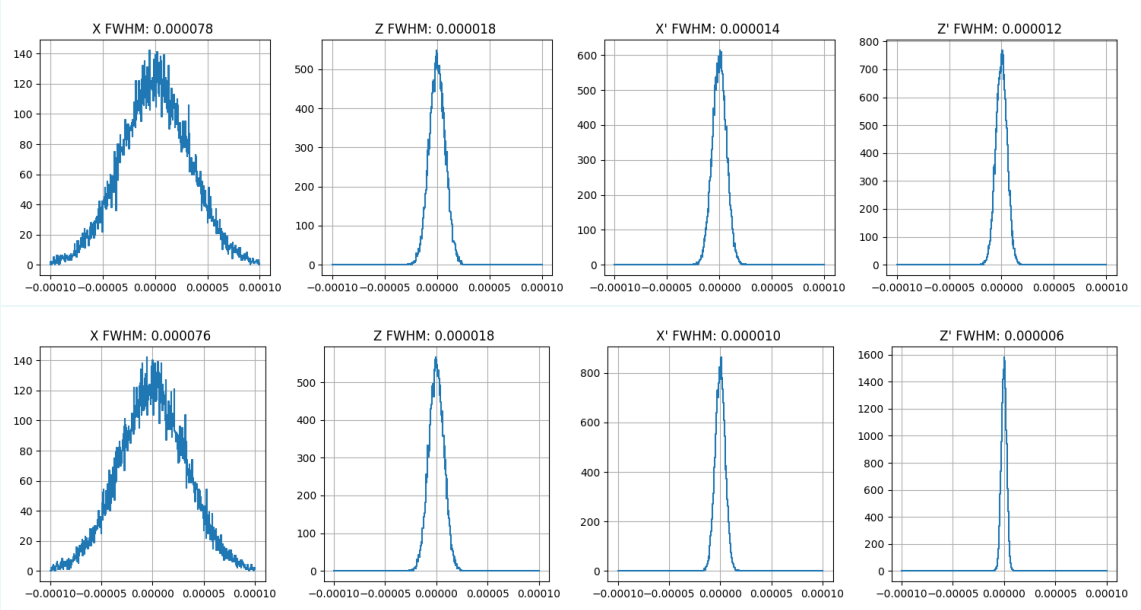


Figure 14: Intensity distribution for real space (X,Z) and divergence space (X', Z') for an undulator (Gaussian approximation) at E_1=15 keV (top) and E_9=135 keV (bottom). No electron energy spread considered.

ii) Activate the "Energy spread" and enter the correct harmonic number under the "Undulator Setting" tab. Observe the differences: significant increase of the divergence at high harmonics.

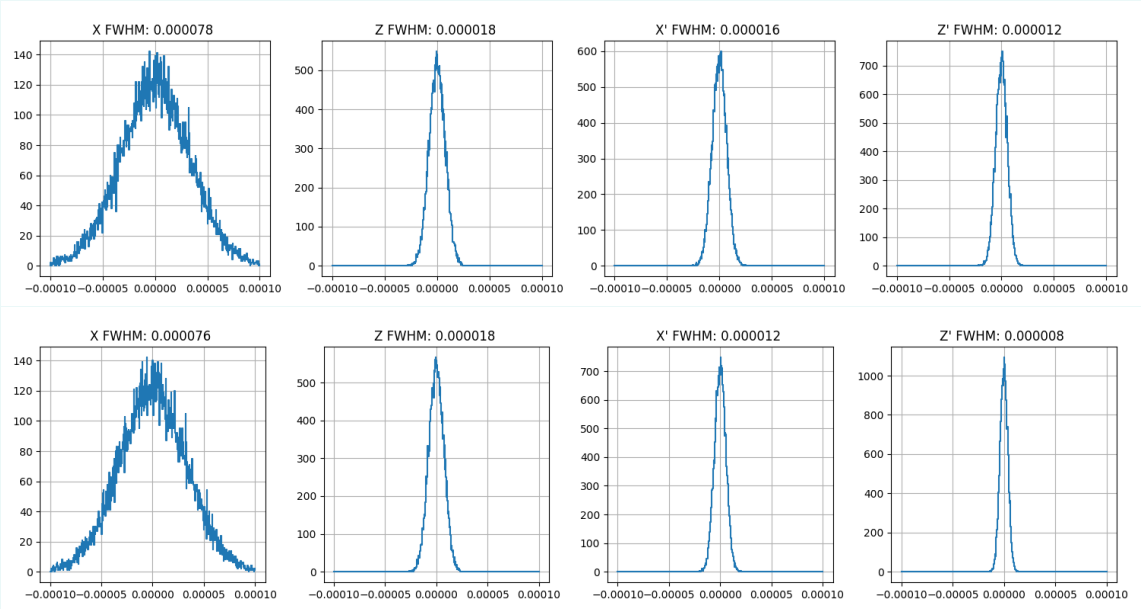


Figure 15: Like figure 14 but with electron energy spread considered. Observe the increase of the divergences for the high harmonics.

iii) The “Undulator Light Source" wiggler require a careful tunning. To start, consider the monochromatic emission at resonance (use “Set to resonance" option). Look at the “Radiation (far field)" under the “Undulator Plots". You see the limits (calculated automatically) are $\theta_{\max} \approx 15$ μrad. Set, for the moment “Advanced Setting" “Size sampling in real space=point". To prepare the setting for off-resonance, select now “Set photon energy=User defined", photon energy E_1=15000 eV and increase $\theta_{\max} = 30$ μrad (see figure 16).

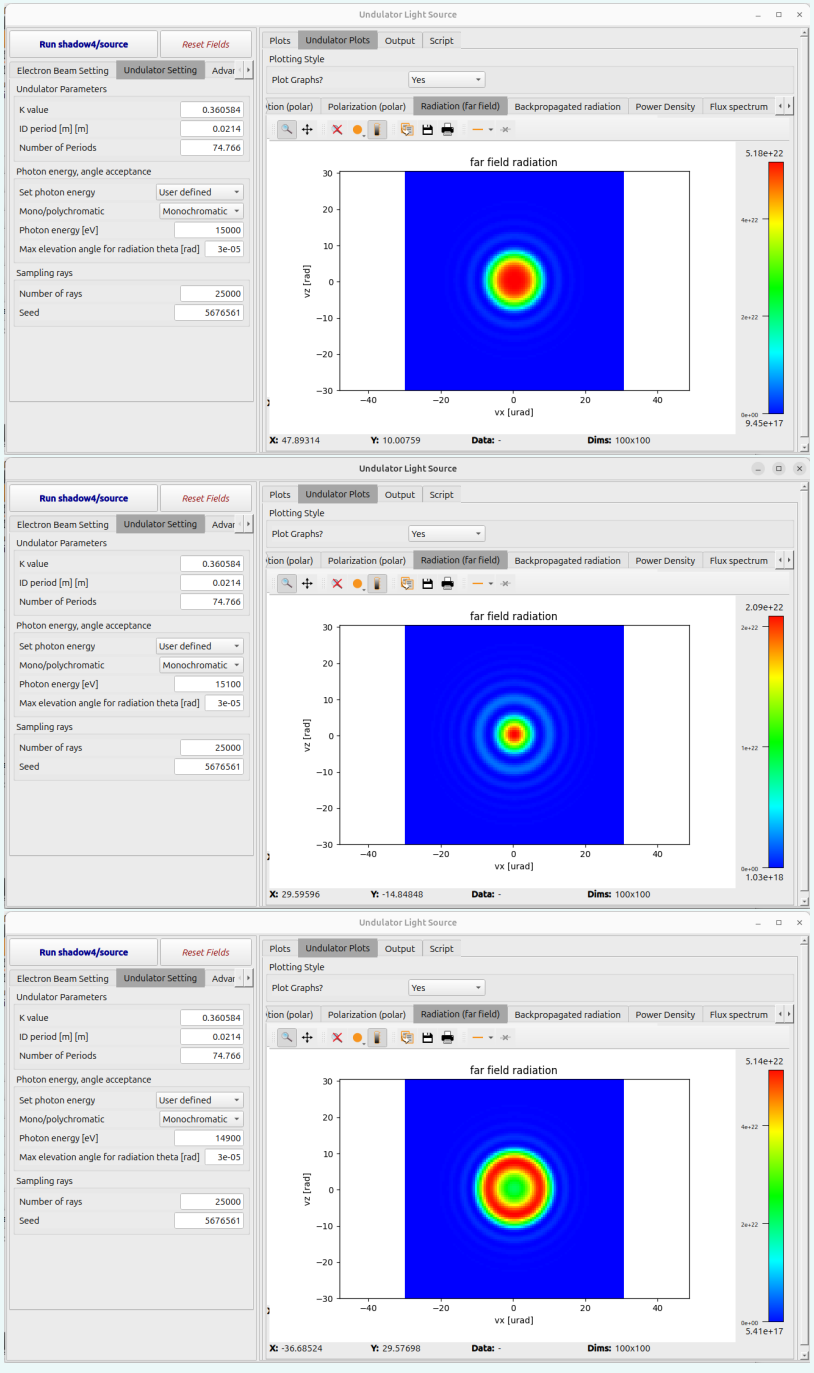


Figure 16: In the “Undulator Light Source" widget, manual set of the far field radiation at resonance E_1 (top), at E_1+100 eV (center), and E_1−100 eV (bottom).

If you want, you can calculate the real size by backpropagating the far field radiation. figure 17 shows an example of how to set the parameters to get “resonable" results.

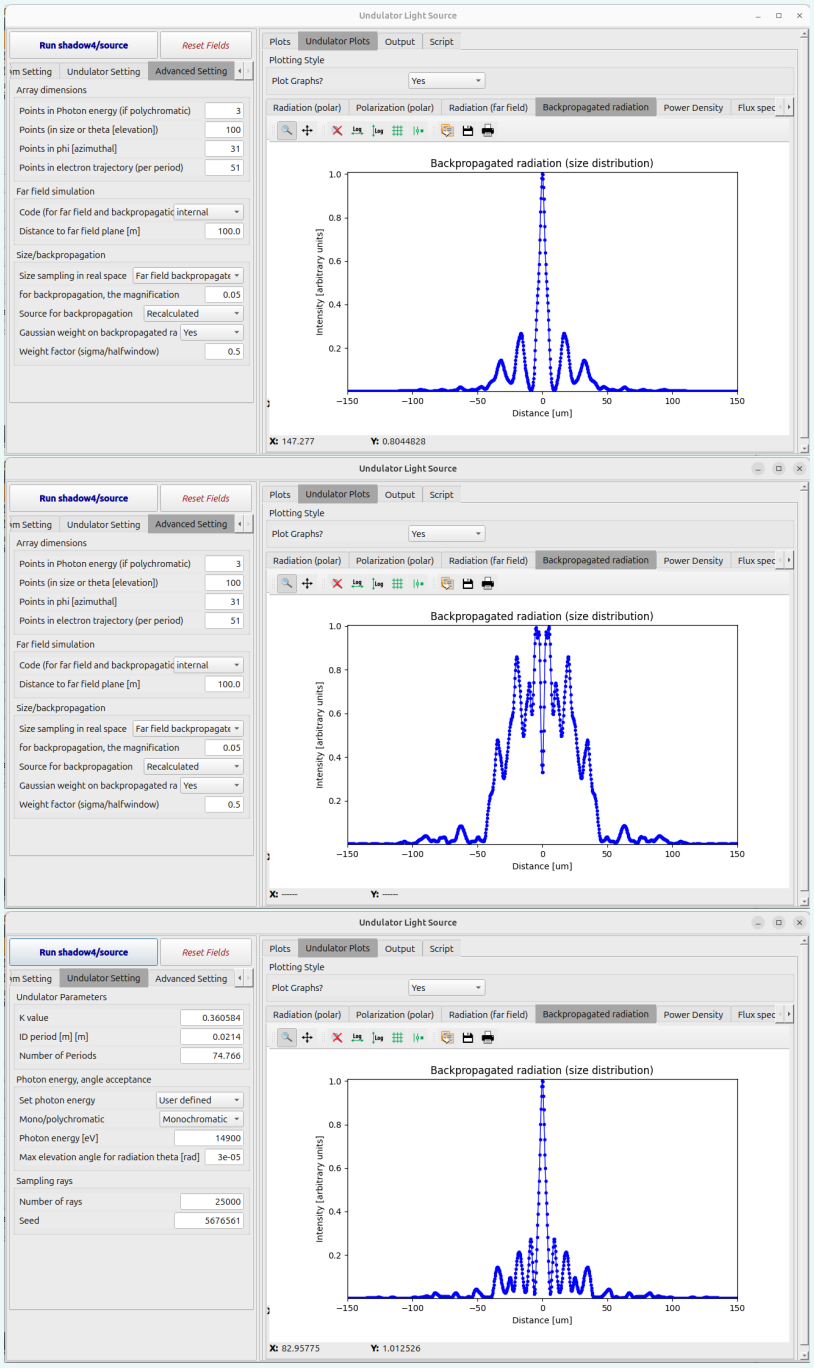


Figure 17: In the “Undulator Light Source" widget, manual set of the backpropagating the far field radiation at resonance E_1 (top), at E_1+100 eV (center), and E_1−100 eV (bottom).

A comparison of the sizes and divergences obtained is in Table 4

Table 4: Sizes and divergences from several “Undulator Light Source" runs.

Energy	**Value**	**Size** μm	**Divergence** μrad
Monochromatic	15000	87.7 × 41.3	13.2 × 13.3
Monochromatic	15100	85.9 × 41.6	13.6 × 7.3
Monochromatic	14900	92.0 × 41.7	17.8 × 16.8
Polychromatic	14900-15100	93.5 × 46.3	30.5 × 36.6

iv) Set the polychromatic option, and in “Advanced Setting" input the wanted number of energy point (the calculation becomes slow!). Results are in Table 4.

6. Beam propagation (ellipses in phase space)

6.1. Phase Space Representation of an Optical Beam

In paraxial (ray) optics, a ray propagating along the y direction can be studied in a transverse direction (x or z, we use x) by two coordinates:

$$\mathbf{r} = \begin{pmatrix} x \\ x' \end{pmatrix}, \tag{20}$$

where

- x is the transverse height of the ray at the current plane, and
- $x' = dx/dy$ is the paraxial angle (slope) with respect to the optical axis.

A *beam* is a bundle of many rays; it is represented as a cloud of points $(x_i,\, x'_i)$ in the phase plane.

A ray travelling a distance d through a uniform medium of refractive index n is transformed by

$$\begin{pmatrix} x \\ x' \end{pmatrix}_{\text{out}} = \underbrace{\begin{pmatrix} 1 & d/n \\ 0 & 1 \end{pmatrix}}_{M_{\text{drift}}} \begin{pmatrix} x \\ x' \end{pmatrix}_{\text{in}}, \tag{21}$$

so explicitly

$$x_{\text{out}} = x_{\text{in}} + \frac{d}{n}\, x'_{\text{in}}, \tag{22}$$

$$x'_{\text{out}} = x'_{\text{in}}. \tag{23}$$

In phase space this is a **horizontal shear**: the angle is unchanged, but each ray's position shifts by an amount proportional to its slope and to the propagation distance. Geometrically, every point in the phase-plane cloud slides to the right by $d\,x'/n$, so the cloud as a whole shears.

Two important limiting cases:

- **Point source** at $x = 0$: the cloud is a vertical line that shears into a tilted line — the beam diverges spatially.
- **Collimated beam** ($x' = 0$ for all rays): the cloud is a horizontal line and is unaffected by the drift — the beam width is preserved.

The ABCD Ray-Transfer Matrix Any *paraxial* optical element can be described by a 2×2 *ray-transfer* (or ABCD) matrix

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \qquad \begin{pmatrix} x \\ x' \end{pmatrix}_{\text{out}} = M \begin{pmatrix} x \\ x' \end{pmatrix}_{\text{in}}. \tag{24}$$

Table 5 lists the matrices for the most common elements.

6.2. Ray tracing simulations

Table 5: ABCD matrices for common paraxial optical elements.

Element	A	B	C	D
Free propagation (distance d, index n)	1	d/n	0	1
Thin lens (focal length f)	1	0	$-1/f$	1
Flat interface ($n_1 \to n_2$)	1	0	0	n_1/n_2
Curved mirror (radius R)	1	0	$-2/R$	1

You will:

- Define a geometrical source with Gaussian profiles.
- Define screens and slits.
- Observe the changes in phase space when the beam propagates.
- Study the effect of a slit or aperture.

Questions:

i) Create a geometric source with Gaussian profiles (in real space: 1 mm × 0.1 mm, in divergence space: 1 mrad × 0.1 mrad; rms values, H × V) and add several screens at 0 (source position), 0.5, 0.75 and 1 m from the source. See the tilt of the (x,x') (horizontal) and (z,z') (vertical) diagrams.

ii) Define an aperture (500 × 50 μm) in the first screen and see its effect in the other screens placed downstream from the beam.

Answer

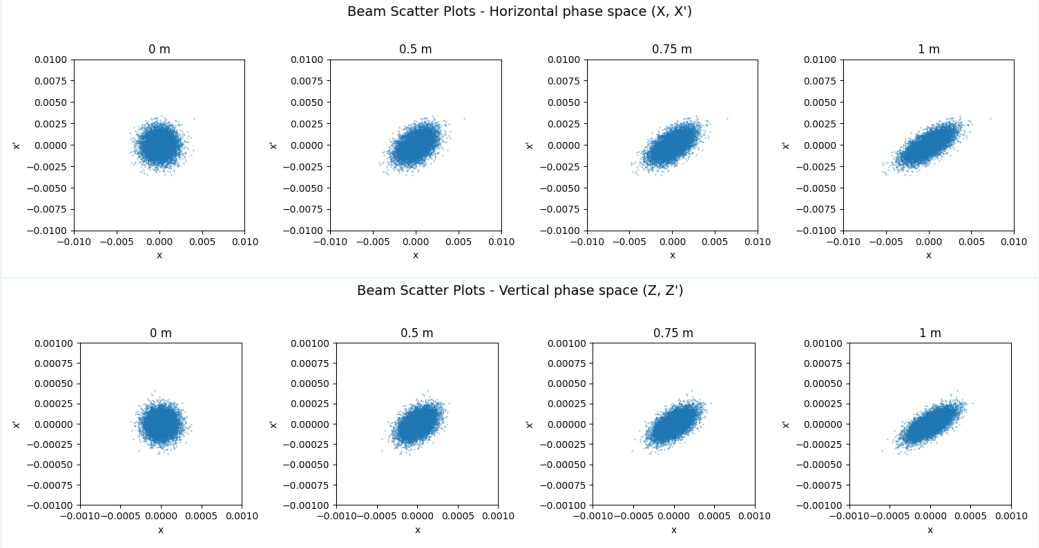


Figure 18: Evolution of the phase space. Top: in X (horizontal), bottom: in Z (vertical).

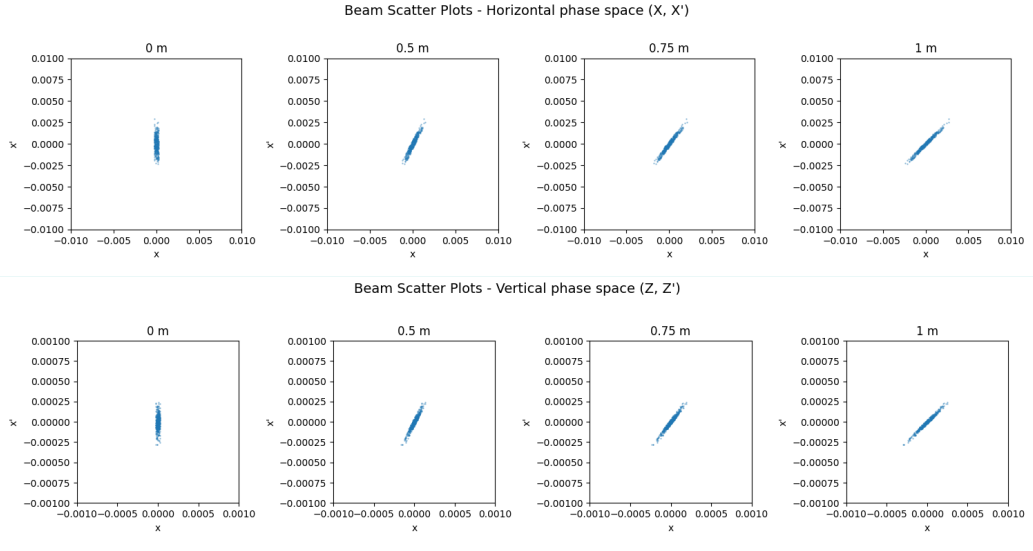


Figure 19: Evolution of the phase space. Effect of a slit in the first screen. Top: in X (horizontal), bottom: in Z (vertical).

Note that instead of defining the different screens, one can directly plug a "Plot XY" to the source and widget and use "Position of the image" = Retraced and play with the distances.

7. Focusing with grazing incidence mirrors: effect of aberrations

7.1. Focusing with grazing incidence mirrors

For a mirror at grazing incidence, let θ be the **glancing angle** (complement of the angle of incidence γ, i.e. $\theta = 90° - \gamma$), R the radius of curvature, and p, q the source-mirror and mirror-object distances.

In the case of normal incidence, the lens equation can be used to calculate the mirror R

$$\frac{1}{p} + \frac{1}{q} = \frac{2}{R} \tag{25}$$

In case of grazing incidence, the R differs if we want to focus in the tangential (or meridional) plane (the plane that contains the source, mirror center and image center), or the sagittal plane (perpendicular to th emeridional one).

Tangential (meridional) plane:

$$\frac{1}{p} + \frac{1}{q} = \frac{2}{R_t \sin\theta} \tag{26}$$

Sagittal plane:

$$\frac{1}{p} + \frac{1}{q} = \frac{2\sin\theta}{R_s} \tag{27}$$

The ratio of the two radii is :

$$\frac{R_s}{R_t} = \sin^2\theta \tag{28}$$

At small glancing angles ($\theta \ll 1$), a spherical mirror $R_t = R_s = R$ will produce a tangential focus is much closer to the mirror than the sagittal focus, leading to large astigmatism.

Toroidal Mirror — Stigmatic Condition A toroidal mirror with tangential radius R_t and sagittal radius R_s can eliminate astigmatism. Since $\sin^2\theta \ll 1$ for grazing incidence, the sagittal radius must be much smaller than the tangential radius: $R_s \ll R_t$.

Mirror shapes for point-to-point focusing Mirrors shaped as elliptical cylinders and as ellipsoids of revolution are obvious choices for many applications as the two-dimensional and three-dimensional surfaces of exact point-to-point imaging.

Paraboloidal cylinders and paraboloids of revolution can be regarded as a special case of an ellipse with one conjugate equal to infinity.

Note that a perfect (on-axis) point-to-point focusing does not mean aberration free. Indeed, the Abbe sine condition is a requirement that a optical system must satisfy to form a stigmatic (aberration-free) image of an off-axis point, not just an on-axis point. It is a known result in Optics that a single reflection surface cannot verify the sine condition. At least two reflections are needed.

If the sine condition is satisfied, the system is free of coma (to all orders) for that object point. If it is violated, off-axis rays focus at different heights than paraxial rays — this is coma. A system that satisfies both the sine condition and forms a perfect axial image (no spherical aberration) is called aplanatic.

Critical Angle for X-ray Reflection: Fresnel Equations For X-rays, the refractive index of matter is slightly less than unity:

$$n = 1 - \delta + i\beta \tag{29}$$

where $\delta \sim 10^{-5}$–10^{-6} is the refractive index decrement and β accounts for absorption. With $n_1 = 1$ (vacuum) and $n_2 = n$, the Fresnel reflection coefficients for s (TE) and p (TM) polarizations, expressed in terms of the glancing angle α, are:

$$r_s = \frac{n_1 \sin\alpha - \sqrt{n_2^2 - n_1^2 \cos^2\alpha}}{n_1 \sin\alpha + \sqrt{n_2^2 - n_1^2 \cos^2\alpha}} \tag{30}$$

$$r_p = \frac{n_2^2 \sin\alpha - \sqrt{n_2^2 - n_1^2 \cos^2\alpha}}{n_2^2 \sin\alpha + \sqrt{n_2^2 - n_1^2 \cos^2\alpha}} \tag{31}$$

Since $n_2 \approx 1$ for X-rays, $r_s \approx r_p$ and polarization is essentially irrelevant. Both reduce to the same expression governed by the quantity:

$$Q \equiv \sqrt{n_2^2 - \cos^2\alpha} = \sqrt{\sin^2\alpha - 2\delta + 2i\beta} \tag{32}$$

where the second form uses $n_2^2 \approx 1 - 2\delta + 2i\beta$ and $n_1 = 1$.

Setting $\beta = 0$, total external reflection occurs when the real part of Q vanishes and the refracted wave becomes evanescent:

$$\sin^2\alpha_c = 2\delta \tag{33}$$

For small glancing angles ($\sin\alpha_c \approx \alpha_c$) this gives:

$$\boxed{\alpha_c = \sqrt{2\delta}} \tag{34}$$

For $\alpha < \alpha_c$, Q is purely imaginary and reflectivity (for non-absorbing mirrors or β=0) is 1 (total reflection).

7.2. Ray tracing simulations

You will:

- Use different mirror shapes in SHADOW
- Experience with the automatic calculation of the mirror parameters
- Include mirror reflectivity

Questions:

Create a geometrical source with Gaussian shape (sx=57 um, sz=10.4 um) and Gaussian divergence (sx'=88.5 urad and sz'=7.2 urad) to simulate the emission of an ESRF 1.65 m undulator at 10 keV in a Low beta section.

i) Study the case of different mirror shapes (spherical, toroidal and ellipsoidal) for focusing the source with distances (p,q)=(30m,10m) (magnification 1/3) and (30m,1m) (magnification 1/30). Set the grazing angle to 2 mrad. Study the effect of the spherical aberrations and its influence depending on the magnification factor. Study the dependence on mirror dimensions and incident angle.

ii) Enter the effect of mirror reflectivity for the Ellipsoidal mirror at M=1/3. Consider a Rh (density=12.4 g/cm3) coating and a source with energy distribution in 5-45 keV (box-distribution). Define mirror finite dimensions 0.2×1 m^2 (center it well, footprint is nor symmetric). Plot also the intensity versus energy.

Hints: The workspace file `aberrations.ows` contains the workflow of this exercise.

Notes: For the different mirrors, shadow4 can calculate the surface parameters (curvature radii, ellipse axes, etc). by selecting the parameters in Basic Settings → Surface Shape → Type = "internal/calculated". The resulting mirror parameters can be seen using the "Info" widget ("OE Info" tab). For including reflectivity, you can define internally using either the "xraylib" or "DABAX" library. Alternatively, you can use the preprocessor "PreRefl" for creating the material file.

Answer

i)

The aberrations effect increases if

- one goes to more grazing angle
- one reduces the magnification factor (i.e., mode demagnification of the source)
- one uses larger mirrors. Small mirrors reduce aberration because cut rays which arrive far from the mirror center. Obviously, this effect reduces also the intensity. For analysing that, use the Basic Settings->Dimensions->Limits Check entry.

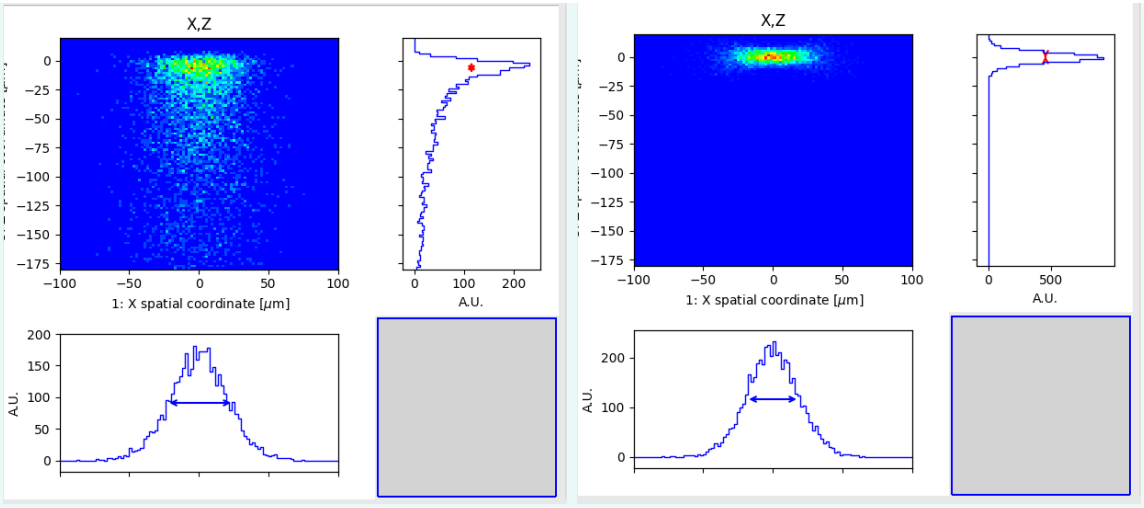


Figure 20: Mirror imaging with magnification M=1/3: Toroidal (left) and Ellipsoidal (right).

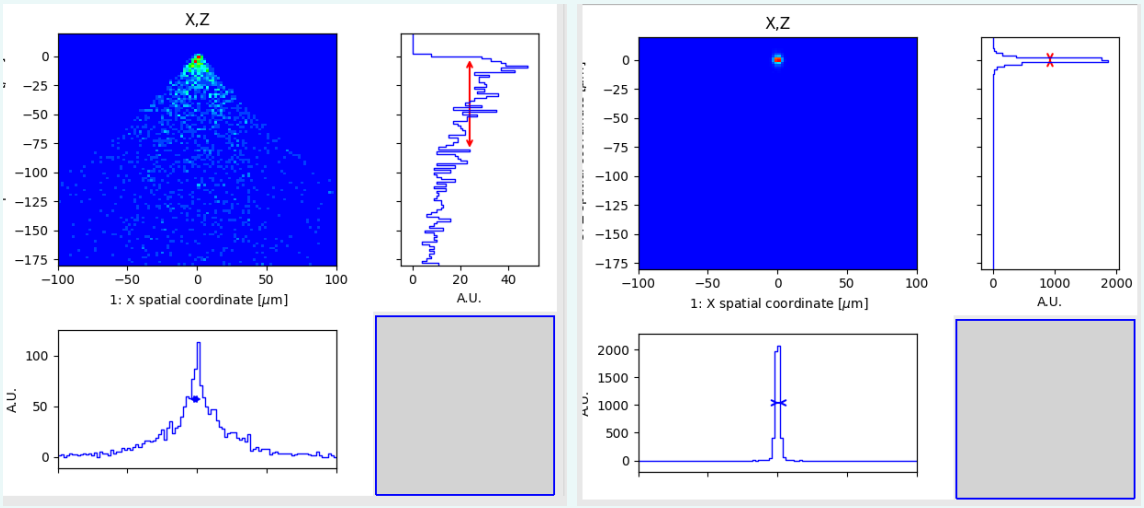


Figure 21: Mirror imaging with magnification M=1/30: Toroidal (left) and Ellipsoidal (right).

ii)

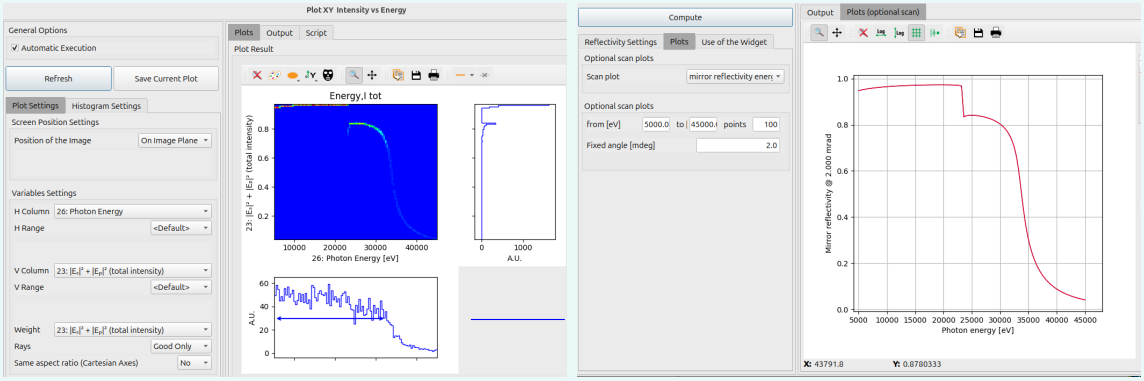


Figure 22: I(E) plot from rays (left) or from PreRefl proprocessor (right).

8. Kirkpatrick-Baez system with slope errors

8.1. Kirkpatrick–Baez (KB) Mirrors

KB mirrors are a system of **two orthogonal grazing-incidence mirrors** arranged in series, each focusing in one plane only. Proposed by Kirkpatrick and Baez (the father of the singer Joan Baez) in 1948, they provide a practical solution to the astigmatism inherent in single grazing-incidence mirrors, and reduce significatively abberations (coma) because the KB system approximates an imaging system that verifies the Abbe sine condition.

As follows from the Coddington equations, a single elliptical mirror at grazing incidence focuses strongly in the tangential plane but weakly in the sagittal plane. The KB solution decouples the two planes:

- **Mirror** M_1 focuses in the vertical (horizontal) plane.
- **Mirror** M_2 focuses in the horizontal (vertical) plane.
- The two mirrors are oriented at 90° to each other.

Each mirror is a section of an ellipse whose two foci coincide with the source and the desired focal point:

$$\text{Source} \longrightarrow \boxed{M_1 \text{ (focus in plane 1)}} \longrightarrow \boxed{M_2 \text{ (focus in plane 2)}} \longrightarrow \text{Focus}$$

The order matters: M_1 and M_2 present different magnifications. Usually in Synchrotron Radiation rings, the emittance (therefore source size) is larger in horizontal, therefore it is interesting to have the HFM (horizontal focusing mirror) closer to the image to profit the higher demagnification.

Focusing Condition Each mirror obeys its the Coddington equations because of the "definition" of ellipse: point-to-point focusing for any point of the ellipse. For mirror M_i with glancing angle θ_i, object distance p_i and image distance q_i:

$$\frac{1}{q_i} + \frac{1}{p_i} = \frac{2}{R_i \sin\theta_i} \qquad \text{(tangential, focusing plane)} \tag{35}$$

Each mirror is designed so that the tangential equation is satisfied exactly for the chosen conjugates p_i and q_i (yielding an elliptical profile). The mirrors are flat in the sagittal direction.

Aberrations On-axis focusing is in principle perfect for each plane independently. Residual aberrations arise from i) Coma: each mirror satisfies the Abbe sine condition only approximately; coma grows with the numerical aperture, and ii) Cross-coupling, because the magnification the two mirrors are different.

Advantages

- **Decoupled adjustment**: each mirror can be tuned independently in its focusing plane.

- **Flexible demagnification**: different conjugate distances can be chosen for each plane, allowing asymmetric focusing (e.g. a round spot from an asymmetric source).
- **Broadband**: no wavelength selectivity; works across a wide photon energy range.
- **Relatively easy fabrication**: each mirror requires only a 1D elliptical figure, far simpler to manufacture and measure than a toroid or full ellipsoid.

Disadvantages

- **Two reflections**: throughput is reduced by absorption and scattering losses on both mirrors, increasingly so at higher photon energies.
- **Alignment sensitivity**: a tilt of M_2 introduces astigmatism; the two mirrors must be carefully co-aligned.
- **Longitudinal separation**: the two focal planes are slightly displaced unless the design explicitly accounts for the separation between M_1 and M_2.

Simulations of the surface irregularities (slope errors) Mirrors used in grazing incidence and making a high demagnification are very sensitive to the surface imperfections. Indeed, the mirrors perfoemance in current synchrotron beamlines is limited by surface errors. It is therefore important to simulate this effect. In SHADOW, this is done overlapping the mirror error profile defined in a mesh file. This contains the deformation heigh surface $z(x, y)$ that is added to the ideal elliptical shape.

We use here the DABAM (DAta BAse of Metrology) database to retrieve the error profiles. It part of the OASYS project. The repository `https://github.com/oasys-kit/DabamFiles/` contains a number of entries, each consisting of two files:

- dabam-NNN.txt — a JSON header with metadata about the mirror
- dabam-NNN.dat — the actual metrology data (slope profile measurements).

They are accesible from OASYS using the “DABAM Heigh Profile" widget (in Syned-Tools) and can be plugged directly in the SHADOW4 widgets.

8.2. Ray tracing simulations

You will:

- define an optical system with two mirrors of elliptical section (Kirkpatrick-Baez system) with high demagnification
- verify that this system does not work with circular mirrors (instead of ellipses)
- configure SHADOW to calculate automatically the ellipse parameters
- learn the impact of the “optical element orientation angle" and use of the “Empty Element" widget to undo the rotations
- study the degradation of the focal spot when introducing surface errors (slope errors) using DABAM

Questions:

Study the case based on the KB at the ESRF-ID21 beamline. It has a large demagnification (the KB is at about 50 m from source and the focal position is at roughly 0.1 m, thus demagnification of $1/M \approx 500$).

The exact position of the elements are in the Tables.

Mirror dimensions: VFM: length=47 mm and width=20 mm. VFM: length=90 mm and width=20 mm. Use 25000 rays. Do not include the mirror reflectivity.

Table 6: Distances for all optical elements.

OE	Type	p [m]	q [m]	src–OE [m]	src–screen [m]
1	SCR/SLIT/ABS	51.0000	0.0000	51.0000	51.0000
2	Mirror	0.5000	0.0575	51.5000	51.5575
3	Mirror	0.0325	0.0600	51.5900	51.6500

Table 7: Focusing elements with design distances and magnification.

OE	Shape	$p_{\rm foc}$ [m]	$q_{\rm foc}$ [m]	$1/M$
2	Elliptical cylinder	51.50	0.15	343.33
3	Elliptical cylinder	51.59	0.06	859.83

Hints: you may use the workspace file `kirkpatrickbaez.ows`.

Answer

In the workspace there are several branches with different options:

The first branch corresponds to the typical KB with elliptical mirrors. Several important aspects must be highlighted:

- The first mirror has α=0 (VFM, facing up), and the second has α=90^{circ} (HFM, facing laterally). As discused in section 2.1, the rotation imply the rotation of the axes, thus, for mirror 2, the display (X,Z) corresponds to (vertica,Horizontal) directions. The "Plot XY" widget, displays columns (3,1) to see the plot as in real life.
- The mirror figure is Ellipsoidal (we start with shape with revolution symmetry around y), but then we select "Cylindrical=Yes" to make a "cylinder", i.e. to flat one direction. The surface is flat is along x (thus the cylinder axis has an angle 0 with x).
- we tell SHADOW to compute the ellipse parameters ("internal/calculated"). The foci and "continuations planes" (CP) are "Different", because the focal points are not in the CPs. The calculated values can be displayed with the "Info" widget, in the OE Info tab.

Second branch corresponds to the KB with circular (cylindrical) mirrors. The aberrations are enormous and cannot be used in real world.

The third branch considers the reflectovity of the mirrors (coated with Ni). It also

shows a “trick" to reverse the axes: use an “Empty Element" without change in the incidence direction (incident angle zero, output angle 180^{circ}) and “undo" the 90 deg rotation, that means setting $\alpha = -90 = 270$ deg. The resulting spot is very small (20 nm $\times$ 20 nm).

The 4th branch shows the effect of slope errors. We use for that the “DABAM" widget available from the SYNED add-on. Select the profile 63 and 62 for the VFM and HFM, respectively. Set a reasonable file name for the output (e.g. `kb_vfm.hdf5` and `kb_hfm.hdf5`). Process clicking sequentially “retrieve profile", “Generate Height Profile File" and connect to the mirror widget. Note that using a mesh file requires setting the mirror dimensions to lengths smaller those sampled in the height profile widget. If dimensions are not set, you will be proposed to accept the one calculated automatically.

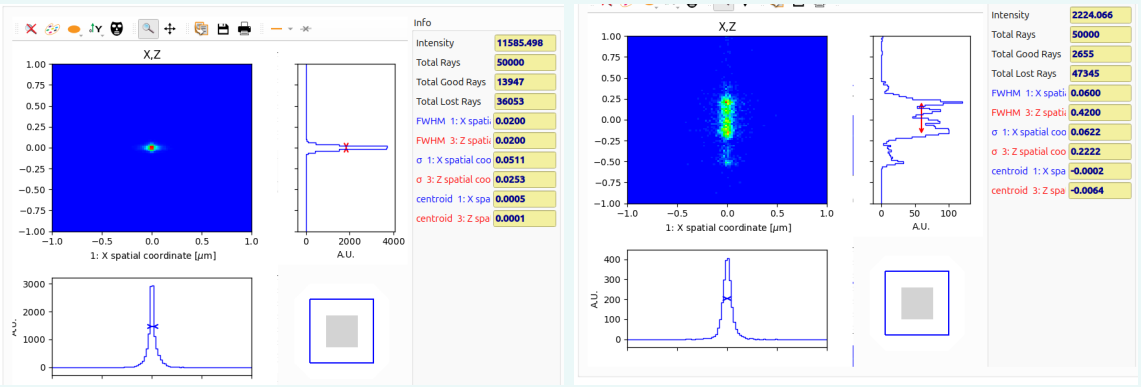


Figure 23: Intensity distribution at the focal plane produced by the KB. Left: ideal surfaces without slope errors. Right: using slope errors from measured data (DABAM library) (0.91 μrad r.m.s. for VFM and 0.51 for HFM).

9. Double crystal monochromator

9.1. Basics of crystal optics

In a perfect crystal, diffraction is governed by **dynamical theory**. Unlike mosaic crystals, a perfect crystal reflects nearly 100% of incident radiation within a specific angular range known as the **Darwin Width** (w_D). This is the intrinsic angular width of the reflection curve for a perfect crystal. It is typically on the order of arcseconds (a few microradians) for hard X-rays.

The reflection profile (or rocking curve) typically exhibits an asymmetric "flat-top" shape.

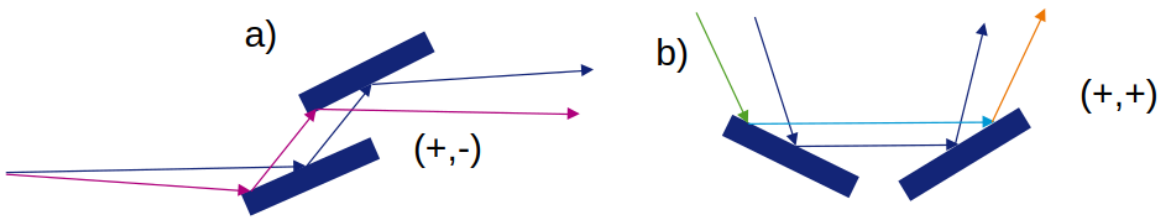
The Double-Crystal Monochromator (DCM) The standard synchrotron setup uses two parallel crystals. This **parallel** $(+,-)$ **geometry** is preferred for two reasons: i) **Beam Stability:** The second crystal restores the beam direction to be parallel to the incident beam, providing a constant vertical offset, and ii) **Harmonic Rejection:** By slightly "detuning" the second crystal (rotating it by a few microradians), higher-order harmonics $(2\lambda, 3\lambda)$ are suppressed because their Darwin widths are significantly narrower than the fundamental.

Configurations: $(+,-)$ **vs.** $(+,+)$ These configurations describe the relative orientation of the crystals and their dispersive power.

Config.	Geometry	Type	Effect
$(+,-)$	Parallel	Non-dispersive	High intensity; energy resolution $\Delta E/E$ is coupled to source divergence.
$(+,+)$	Anti-parallel	Dispersive	High energy resolution; $\Delta E/E$ is decoupled from source divergence but intensity is lower.

Table 8: Comparison of Monochromator Configurations

Figure 24: Double crystal configurations a) (+,-) where all rays accepted by the first crystal are also accepted (same angle) at the second crystal; b) (+,+) where the out of axis rays on the first crystal are incident at different angles on the second crystal.



The DuMond Diagram and the Source The DuMond diagram plots wavelength (λ) against angle (θ). The source is represented as a broad rectangle defined by its spectral range ($\Delta\lambda_s$) and angular divergence ($\Delta\theta_s$).

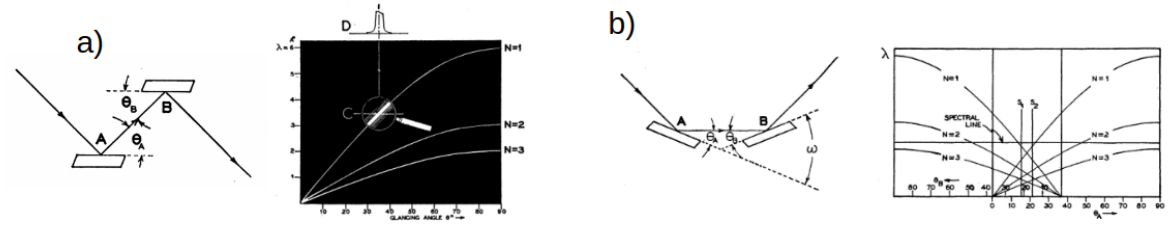
In the $(+,-)$ case, the Bragg bands for both crystals have the same slope:

$$\frac{d\lambda}{d\theta} = 2d\cos\theta \tag{36}$$

Because the slopes are identical, they overlap over a large area within the source window. This results in high flux but makes the energy resolution dependent on the source's angular spread.

In the $(+,+)$ case, the second crystal has a negative slope relative to the first. The two bands intersect like an "X." The resulting monochromatic beam is defined only by the tiny diamond-shaped intersection of the two bands. This configuration effectively "ignores" source divergence to produce a highly monochromatic beam.

Figure 25: Du Mond diabgrams for a) (+,-) configuration, and b) (+,+) configuration. J. W. M., DuMond, Phys. Rev. 52, 1937, 872-883.



9.2. Ray tracing simulations

You will:

- define crystals in SHADOW
- use the setting facility to automatically align the crystal
- calculate the energy resolution for a crystal and a combination of systems
- optimise the source bandwidth
- play with the mirror orientation angle. Relate its values to the crystal dispersion ((+,-) and (+,+) crystal combination)
- create crystal reflectivity data and display diffraction profiles using the "bragg" preprocessor

Questions:

Create an undulator source (starting from section **??**) at 5000 eV with a "large enough" energy bandwidth (e.g. 5000±5.

i) Implement a flat Si 111 crystal at 40 m from the source. Verify the energy dependence and calculate resolution. Redefine the source energy bandwidth to optimize the calculation in order to obtain the energy dependence with the highest signal.

ii) Add a second crystal 50 mm downstream from the first one in (+,-) and (+,+) configurations (play with the mirror orientation angle).

iii) Repeat the calculation at third harmonic (E=15 keV, crystal reflection 333). Explain the differences in energy resolution.

iv Use the Bragg preprocessor to create the reflectivity data for a Si 111 crystal.

Hints: you may use the workspace file `crystalmonochromators.ows`.

Answer

i) The wide energy range selected is 5000±5 eV. The Energy plot after the first crystal has FWHM 0.7 eV. Therefore, for avoiding to calculate so many rays that are not diffracted, we optimize the source bandwidth to 5000±1 eV. As a rule of thump, for Si 111 we expect an intrinsic resolution $\Delta E/E \approx 10^{-4}$, therefore ΔE=0.5 eV.

ii) For (+,-) (non-dispersive configuration). the optical element orientation angles are 0 and 180 deg for the first and second crystals, respectively. For (+,+) (dispersive configuration) the optical element orientation angle is 0 for both crystals.

We obtain from the histograms 0.7 eV after the first crystal. After the second crystal 0.6 eV for (+,-) and 0.5 eV for (+,+). The dispersive configuration (+,+) gives the intrinsic resolution, and the non-dispersive configuration (+,-) gives a larger bandwidth due to the effect of the beam divergence (that is small, the beam divergence is ≈13 μrad compared with the Darwin width ≈42 μ rad).

iii) For the third harmonic, the energy bandwidth for the (+,-) configuration is 0.36 eV and for (+,+) is 0.06 eV. A large difference is observed here, because the beam divergence (12 μrad) is large with respect to the Darwin width (about 4 μrad).

iv) to compute the diffraction profile one can use the Bragg preprocessor. It creates the material data for the crystal. You can plug it into the crystal widget to use this file, instead performing the direct calculation. The output file should be created for a large range of energy (e.g., from 4000 to 20000 eV). This widget can also be used to display the diffraction profile at 5000 eV. Alternatively, you can use the XOPPY/CRYSTAL widget.

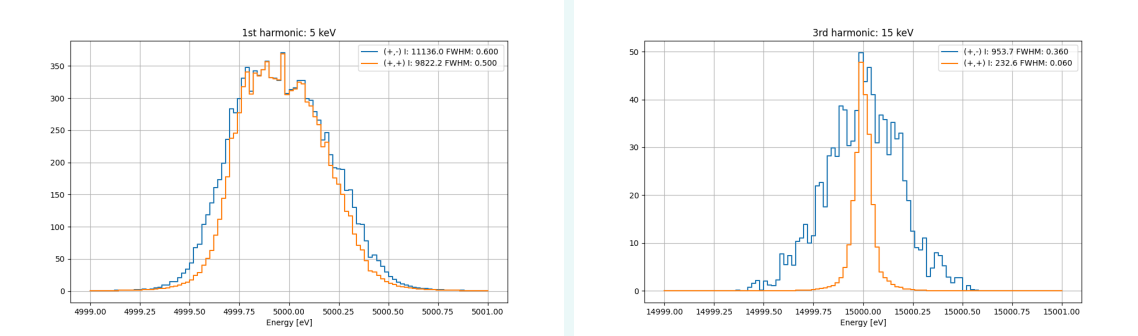


Figure 26: Intensity vs photon energy plots for the two configurations (+,-) and (+,+). Left: 1st harmonic around 5 keV. Right: 3rd harmonic around 15 keV.

10. Sagittal focusing

10.1. Crystal monochromator with sagittal focusing

Bending the second crystal in a double-crystal monochromator is an efficient method of increasing the photon flux density at bending-magnet and wiggler beamlines [19, 20]. In this scheme, sagittal focusing is applied at the second crystal, whose radius of curvature is given by the Coddington equation:

$$R_s = \frac{2\,p\,q\sin\theta_B}{p+q} \tag{37}$$

where p and q are the source-to-crystal and crystal-to-focus distances, respectively, and θ_B is the Bragg angle corresponding to the selected photon wavelength λ, related to the crystal d-spacing d by Bragg's law ($\lambda = 2d\sin\theta_B$).

Despite its efficiency, sagittal focusing with a bent crystal presents several practical challenges:

- **Variable curvature requirement.** The required radius of curvature changes with the Bragg angle, necessitating a dynamical bender capable of continuously adjusting R_s during an energy scan.
- **Extreme bending radii.** The required radii can range from approximately 1 to 5 m, placing stringent demands on the mechanical design and stability of the bender.
- **Angular mismatch between crystals.** Sagittal bending introduces a mismatch in the ray angles seen by the first and second crystals, reducing the effective reflectivity. This loss can be mitigated by optimizing the position of the monochromator along the beamline. It has been shown that a demagnification ratio of $p : q = 3 : 1$ minimizes the angular mismatch and maximizes reflectivity [19].
- **Anticlastic bending.** Bending the crystal in the sagittal direction induces a parasitic curvature in the tangential (diffraction) direction, known as anticlastic bending. This introduces aberrations and reduces the focusing efficiency. A common mitigation strategy is to machine a set of ribs into the crystal, whose dimensions and spacing are optimized for the target configuration. However, the ribs can also produce a segmented sagittal profile that deviates from the ideal cylindrical shape, introducing additional aberrations.

Pure bending of even thin rectangular plates (thickness $\ll$ radius of curvature R_s) results in a transverse curvature of the plate with radius R_m by [20]

$$R_m = -\frac{R_s}{\sigma_P}, \tag{38}$$

with σ_P the Poisson ratio (≈ 0.21 for isotropic silicon).

10.2. Ray tracing simulations

You will:

- define a cylindrical mirror for sagittal focusing

- define "externally" the optical element radius of curvature
- verify the best reflectivity is pbtained at demagnification 3
- include anticlastic bending

Questions:

i) Define a bending magnet source (e.q. ESRF1: E_e=6 GeV, I=0.2 A, σ_x=395μm, σ_z=36.9μm,σ'_x=9.87μrad,σ'_z=1.057μrad, horizontal acceptance 5 mrad). Set the photon energy interval in E=2000±10 eV. Define a (+,-) flat-flat Si111 double crystal monochromator tuned at 20 keV. Duplicate the second crystal and introduce the sagittal curvature to focus in the horizontal plane at the sample position. p=30 m, q=10 m. Calculate and compare the horizontal spot size.

ii) With the help of a script, calculate the intensity vs magnification factor M. Fix p=30 m and set $q = Mp$. Verify that ratio $M = 1/3$ is the optimum.

iii) Study the anticlastic curvatura. For that, use the toroidal shape, with Major radius the anticlastic, and minor radius the sagittal. To express that R_m has opposite cuvature than R_s, the torus pole must be located at the "upper/inner=convex/concave" location. Try several anticlastic radius, starting from infinity (a large value) and going down to $R_m \approx 7.5$ m where the diffracted intensity is almost zero.

Hints: You may use the workspace files `sagittalfocusing.ows` that also includes the python script that scans the magnification. One can see that the intensity peaks at about 1/3. A script is included to calculate the curvature radius: Rs (20 keV, M=1/3)=148.3 cm.

Answer

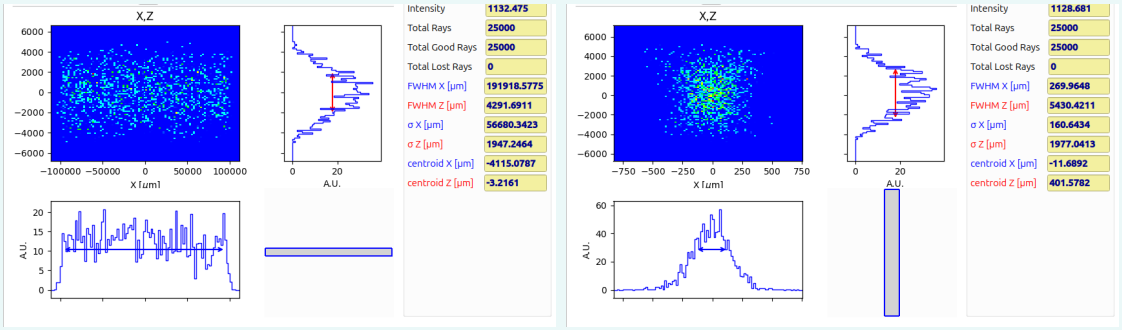


Figure 27: Image at focal plane by the flat-flat system (left) and the flat-sagittal system (right)

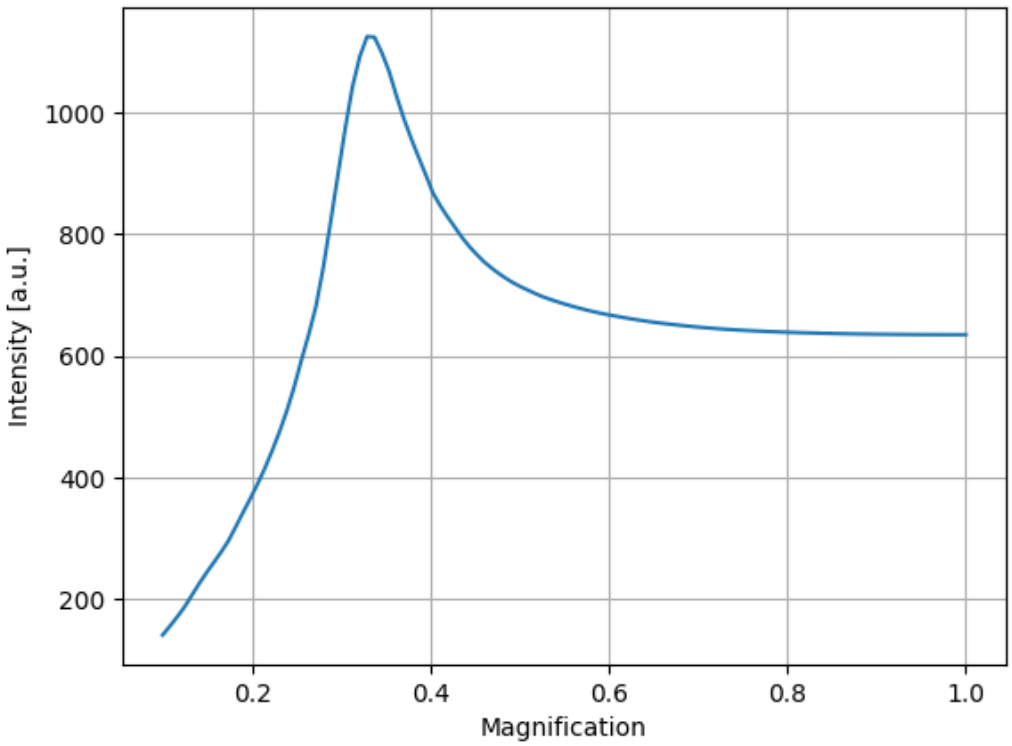


Figure 28: Intensity (in arbitrary units) versus magnification factor M for a point and monochromatic (E= 20 keV) source placed at 30 m from the sagittaly bent crystals. The beam divergences is 5 mrad. We clearly observe the maximum of the transmission at M=0.33, as predicted by the theory.

11. Thermal deformations

11.1. Deformations due to heat load

Synchrotron radiation sources produce extremely intense beams of photons across a broad spectrum, and managing the **heat load** deposited by this radiation is one of the central engineering challenges in beamline design.

Heat load refers to the power deposited by the synchrotron radiation beam onto optical components such as masks, slits, absorbers, monochromators, and mirrors. Modern storage rings and especially fourth-generation sources (such as EBS at ESRF, APS-U, or SPring-8-II) produce beams with total powers ranging from **hundreds of watts to tens of kilowatts**, concentrated into very small footprints.

The critical parameter is not just total power but also the peak power density: the power per unit area that strikes an optical surface.

Overview of the Simulation Loop The simulation pipeline follows this chain:

- Source power spectrum calculation OASYS/XOPPY
- Profile of incident power on the optical element (taken into account the effect of the elements upstream of it)
- Temperature field (FEA)
- Surface deformation (FEA)
- Extract main parameters of deformation (e.q. maximum slope error)
- Iteration and optimization of the main parameters (cooling, dimensions, shapes) using FEA
- Extract FEA-calculated surface for ray tracing verification of the optical effect of the residual deformations
- Optical output (flux, focus, resolution)

In this chapter we study for a monochromator crystal the broadening of the energy distibution due to the surface deformation and the image spot profile degraded by thermal deformation.

11.2. Ray tracing simulations

You will:

- use a python script to create a file sampling a thermal bump
- see the effect of the bump in image and energy resolution.

Questions:

Load the `thermaldeformation.ows` workspace. Run the script to create a Gaussian bump bump.h5 Run the crystal system (a single Si111 crystal) without and with

thermal bump. See the changes in the image and energy resolution.

Answer

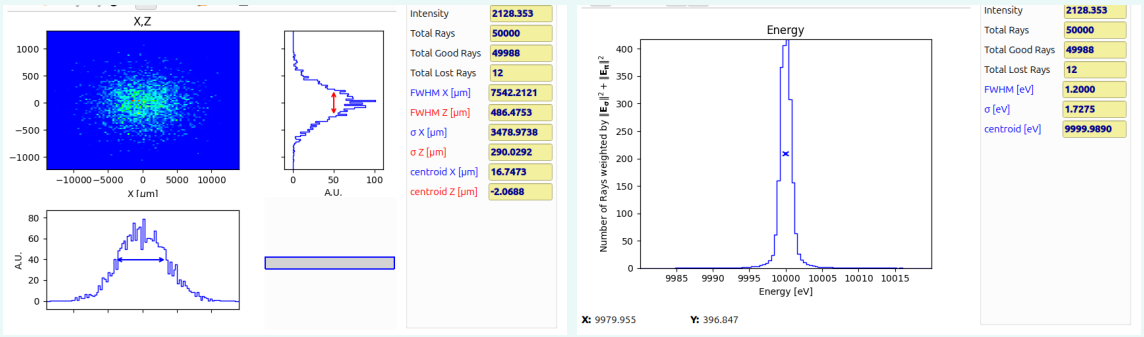


Figure 29: No bump: H: 7.5 mm, V: 0.49 m, ΔE=1.2 eV

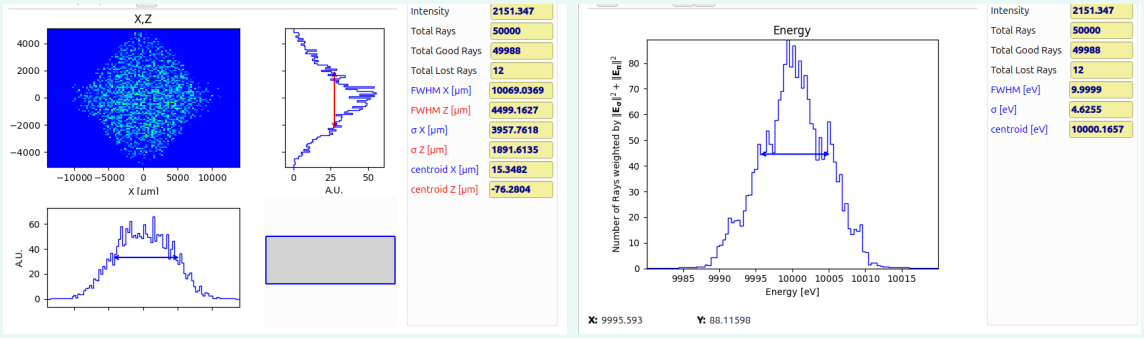


Figure 30: With bump: H: 10 mm, V: 4.5 mm, ΔE=10 eV

12. Curved crystals: Rowland and off-Rowland configurations

12.1. Efficient focusing with curved crystals: Johann geometry and Rowland circle

Consider a crystal bent to radius R, with diffraction planes **parallel to the crystal surface** (symmetric cut, asymmetry angle $\chi = 0$). A point source S at distance p from the crystal centre emits a divergent beam that impinges on the concave surface. The condition on p and the image distance q such that the reflected rays converge to a focus is given by the Coddington equation (for the meridional or diffraction plane)

$$\frac{1}{p} + \frac{1}{q} = \frac{2}{R \sin\theta_B} \tag{39}$$

where
$theta_B$ is the Bragg angle at the crystal centre (it is focused independently in the sagittal plane at a different distance, but this is not discussed here).

The **Johann geometry** bends a flat crystal to radius R and places source S and detector D symmetrically at the *equal-conjugate* distance (i.e. 1:1 magnification) satisfying Eq. (39):

$$p = q = R \sin\theta_B. \tag{40}$$

At the crystal *centre*, this placement is exact: the incidence angle equals
$theta_B$ and the Bragg condition $\lambda = 2d \sin$
$theta_B$ is perfectly satisfied.

Origin of the Johann Error: Now consider a ray hitting the crystal at a point P displaced by a distance ℓ along the crystal surface from the centre. Because the crystal is *cylindrically* bent, its surface follows a circle of radius R. The local surface normal at P is rotated by an angle

$$\delta\phi = \frac{\ell}{R} \tag{41}$$

relative to the central normal. The source S, however, is still at the fixed position defined by Eq. (40). The actual incidence angle at P therefore deviates from
$theta_B$, and the Bragg condition is no longer perfectly satisfied.

Working out the geometry to leading order in ℓ/R, the **angular error** at P is:

$$\delta\theta \approx \frac{\ell^2 \cos^2\theta_B}{2R^2 \sin\theta_B}. \tag{42}$$

This error is *second order* in ℓ: it vanishes at the crystal centre and grows towards the edges. Via the differential of Bragg's law $\delta E/E = -\cot\theta_B\, \delta\theta$, the corresponding relative energy broadening is:

$$\delta E \;\approx\; \frac{\ell^2 \cos^2\theta_B}{2R^2} \tag{43}$$

This is the **Johann error**. Its salient properties are:

- It scales as ℓ^2, so using a smaller crystal reduces the error quadratically, at the cost of collected solid angle.
- It vanishes at $theta_B = 90$ř (backscattering), since $\cos 90$ř $= 0$.
- For a given crystal size ℓ, larger bending radius R reduces the error — but also reduces the collected solid angle.
- It is an *intrinsic geometrical aberration* of cylindrical bending: the crystal surface cannot conform simultaneously to the Bragg condition at every point.

The Rowland Circle Construction. The three defining elements are:

- **Source** S, **crystal centre** C, and **detector** D all lie on the Rowland circle of radius $R_W = R/2$.
- The crystal is tangent to the Rowland circle at C, and bent to radius $R = 2R_W$.
- The chord SD subtends an angle 2 $theta_B$ at the centre of the Rowland circle, consistent with Bragg's law.

As energy (and hence $theta_B$) changes, the focus point D moves continuously along the Rowland circle. This is the basis for **energy scanning in Johann spectrometers** without translating the source.

Extension to the Asymmetric Case In the **asymmetric Bragg geometry**, the diffraction planes make an angle χ (the *asymmetry angle*) with the crystal surface. The incidence and exit angles with respect to the *surface* are then:

$$\alpha = \theta_B - \chi, \qquad \beta = \theta_B + \chi. \tag{44}$$

(Signs are for the beam-compressing case; they reverse for beam expansion.) Both α and β satisfy the Bragg condition with respect to the diffracting planes, since the *planes* are still at angle θ_B to the incident beam.

The asymmetry introduces a **beam expansion factor**:

$$b = -\frac{\sin\alpha}{\sin\beta} = -\frac{\sin(\theta_B - \chi)}{\sin(\theta_B + \chi)}. \tag{45}$$

Here $|b| < 1$ corresponds to angular expansion (spatial compression) and $|b| > 1$ to angular compression (spatial expansion).

Applying Fermat's principle to the asymmetric geometry — where the incidence and exit angles on the crystal surface are α and β respectively — modifies the Coddington equation:

$$\frac{\sin^2\alpha}{p} + \frac{\sin^2\beta}{q} = \frac{\sin\alpha + \sin\beta}{R} \tag{46}$$

For perfect meridional focusing from a point source, the equal-conjugate condition generalises to:

$$p = \frac{2R\sin^2\alpha}{\sin\alpha + \sin\beta}, \qquad q = \frac{2R\sin^2\beta}{\sin\alpha + \sin\beta}. \tag{47}$$

The source and detector are now placed at *different* distances from the crystal, but both still lie on the Rowland circle. To see this, note that the Rowland circle radius is unchanged:

$$R_W = \frac{R}{2} \tag{48}$$

The Rowland circle radius is **independent of asymmetry**. What changes is the angular position of S and D on the circle: S sits at arc angle α and D at arc angle β from the crystal normal, so the crystal is no longer at the midpoint of the SD arc. The inscribed angle theorem still guarantees that the local Bragg condition is satisfied at every surface point that lies on the Rowland circle (Johansson case), or approximately so for the Johann approximation.

12.2. Ray tracing simulations

You will:

- understand the effect of crystal radius in energy resolution and focusing conditions
- calculate the focusing conditions in and out Rowland configuration
- understand the importance of using contour curves with PlotXY simulate an asymmetric crystal

Questions:

i) Using a Gaussian source (σ_x=10μm, σ_z=10μm, σ'_x=10μrad, σ'_z=10μrad), calculate the radius of curvature for the focusing conditions $p = q$ =30 m for a symmetrical Si111 Bragg crystal at 10 keV. Calculate *DeltaE*. This was the Rowland condition. Calculate now *DeltaE* for R=50 m and R=25 m (both off-Rowland). Explain the differences.

ii) Calculate the Rowland conditions for 10 keV, Si111, p=30m and asymmetry angle α=5 deg. Calculate energy resolution and spot size.

Hints: use `rowland.ows`

Answer

i) The radius of curvature from the Coddington equations is R=151.733 m (calculated by the script). It is implemented the first branch, giving ΔE=1.32 eV and focus at 30 m (verified using the caustic widget) of 1.32 μm.

For R=50 m (second branch) we then have ΔE=2.6 eV and the best focus is found using the caustic widget at q=5.95 m and where size is 5.5 μm.

For R=25 m (third branch) we then have $\Delta E > 5$ eV and the best focus is found using the caustic widget at q=2.7 m and where size is 2.24 μm.

ii) Using the script to calculate Rowland conditions, we get R=10.623 mm, q=11.848 m; We get ΔE=0.78 eV; spot size of 170 μm found at $q \approx$6 m.

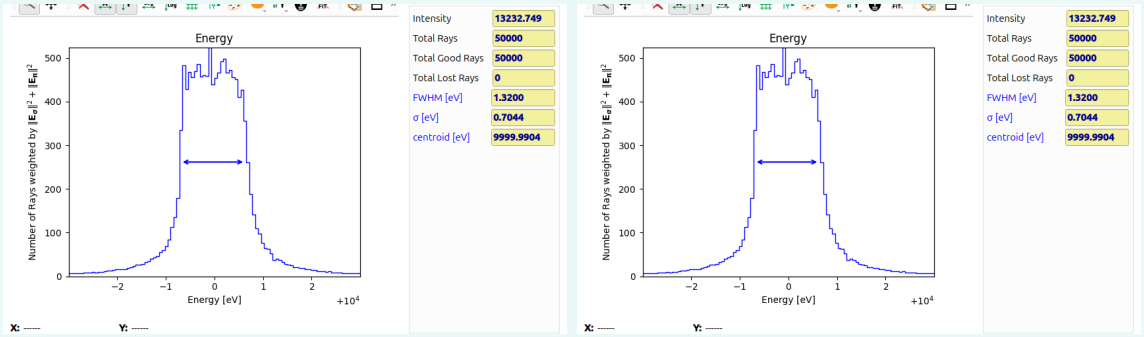


Figure 31: Bandwidth and image at best focus for Rowland configurations.

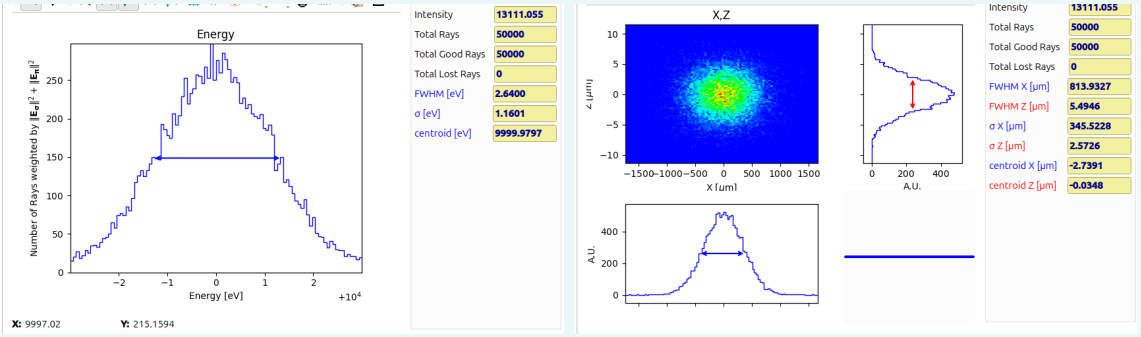


Figure 32: Bandwidth and image at best focus for off-Rowland (R=50 m) configuration.

13. Refractors: X-ray lenses, CRLs and Transfocators

13.1. X-ray lenses

In visible-light optics the refractive index of a material is greater than unity ($n > 1$), meaning light slows down inside the medium and bends toward the surface normal at an interface. For X-rays the situation is reversed: the refractive index is slightly *less* than unity and is written as

$$n = 1 - \delta + i\beta, \tag{49}$$

where δ is the **refractive index decrement** (a small positive number, typically 10^{-5}–10^{-7}) and β is the **absorption index**. Because δ is so small, refraction effects are extremely weak—a single interface deflects an X-ray beam by only micro-radians. This is why refractive lenses for X-rays were considered impractical for so long and why focusing by mirrors (total external reflection), zone plates, and bent crystals was preferred instead.

Two important consequences follow directly from $n < 1$:

- A **concave** lens acts as a converging lens for X-rays (the opposite of visible light), because the beam bends away from the normal at an interface going from a denser to a less-dense medium.
- There is a **critical angle** for total external reflection: for incidence angles shallower than $\theta_c \approx \sqrt{2\delta}$, X-rays are totally reflected rather than transmitted. This limits the maximum useful numerical aperture of a refractive lens.

The Compound Refractive Lens (CRL) Because δ is so small, a single concave lens has an impractically long focal length. For aluminium at 14 keV, $\delta \approx 2.8 \times 10^{-6}$, giving a single-lens focal length

$$F = \frac{R}{2\delta} \approx 54\,\mathrm{m} \tag{50}$$

for a hole radius $R = 0.3$ mm. The solution proposed by Snigirev et al. citeSnigirev1996 is the **compound refractive lens (CRL)**: an array of N identical concave lenses aligned along a common optical axis. The focal length of the compound system is

$$F = \frac{R}{2N\delta}, \tag{51}$$

so stacking 30 lenses reduces the focal length to a manageable ~ 1.8 m.

The diffraction-limited resolution of the lens is

$$\sigma_f = \frac{\lambda\, q}{A}, \tag{52}$$

where A is the effective aperture and λ is the X-ray wavelength. Note that the diffraction limited size is not considered by ray tracing.

Lens Surface Shapes and Aberrations The choice of surface profile has a strong effect on focus quality, especially at high numerical aperture. Spherical lenses are the easiest to manufacture but suffer from severe **spherical aberration**: marginal rays (passing through the lens edge) are focused at a different position from paraxial rays (passing near the optical axis), producing a large, diffuse focal spot. X-ray spherical lenses are no longer

used in practice. Parabolic lenses eliminate most spherical aberration and have been the standard choice since the late 1990s. A parabolic profile $z^2 = 2Ry$ closely approximates the ideal shape for most beamline configurations, and parabolic CRLs routinely achieve micrometre-scale focus sizes. However, the parabola is not a perfect focusing surface.

Elliptical lenses are the theoretically ideal surface for focusing a **collimated** beam to a point. The ellipse is a conic section obtained as a special case of the Cartesian oval when the source is at infinity. Ray-tracing simulations [**?**] show that for nanofocusing applications, elliptical profiles outperform parabolic ones not in terms of the FWHM of the central peak, but in the **intensity of the tails** surrounding it. Those tails reduce the peak intensity (and therefore the gain), so at large apertures the ellipse is clearly preferred.

The **Cartesian oval** is the general ideal surface shape for **point-to-point focusing** from a source at finite distance. The Cartesian oval is a bi-circular quartic curve and reduces to an ellipse (for $n_1 < n_2$, as in X-ray optics) or hyperbola (for $n_1 > n_2$, as in visible-light glass optics) when the source is collimated. In high-demagnification nanofocusing geometries, the Cartesian oval produces a clean Gaussian focal spot with no tails, while elliptical lenses already show some tails and parabolic lenses show significant ones. Table 9 summarises the ideal surface choice depending on beam and focusing geometry for X-ray optics ($n < 1$).

Table 9: Ideal refractive surface shape for different focusing conditions (X-rays, $n < 1$).

Input beam	**Output beam**	**Ideal surface**
Collimated	Convergent (focused)	Ellipse
Convergent	Collimated	Hyperbola
Divergent	Convergent	Cartesian oval

SHADOW works for any x-ray lens simulation, and proceeds as follows.

- Each ray is propagated in a straight line from the source to the first lens surface. The coordinates and direction cosines of the ray at the intersection with the surface are calculated analytically (for simple shapes like paraboloids) or numerically (for Cartesian ovals, using a pre-computed surface mesh.
- At every air–material interface, the Snell's law in vector form is applied. This is repeated at every interface of every lens in the array.
- After traversing a path length l_i through the lens material, the electric vectors of the i-th ray are affected by absorption $e^{-\mu_i l_i}$, where μ_i is the linear absorption coefficient at the ray's energy, including photoelectric absorption and Compton scattering.

SHADOW can output the full two-dimensional intensity distribution at any plane, from which one extracts the focused spot size, the effective aperture, or compute chromatic aberrations under polychromatic illumination. Obviously, the effect of upstream optics such as monochromators or mirrors can be included in the simulations.

The main limitation of geometric ray tracing is the absence of diffraction. For very small focal spots (below $\sim 100\,\text{nm}$) or high-NA configurations, diffraction broadening becomes comparable to the geometric spot size and must be added separately, typically by convolving the ray-tracing point-spread function with an Airy disc of diameter $\sim \lambda/\text{NA}$. The "hybrid" method, that corrects the ray tracing of some deffractive effects is available in OASYS and can be used for that.

13.2. Ray tracing simulations

You will:

- Learn to simulate CRLs and transfocators
- Find the position of the best focus
- Study the changes due to the source energy (chromatic aberrations)
- Define a lens with elliptical shape

Questions:

i) Calculate the first working CRL [18].

ii) Implement a transfocator consisting of two CRLs, 2D-focusing (in both H and V), the first made in Be (16 lenses of radius 200 μm, separated 50μm), and the second in Al (21 lenses of radius 200 μm, separated 50μm). The focal distances are p=31.50 m and q=10 m. This setup is discussed in [1].

Implement a monochromatoc Gaussian source of 48.2 (H) x 9.5 (V) μm RMS size and 100 (H) x 4.3 (V) μrad RMS divergence. Create the same source for two different energies (E1=35200 eV and E2=35700 eV).

Study the beam evolution close to the focal plane using the “caustic" widget. You may find a small astigmatism, i.e., the H and V foci are at slightly different position (you will find that the astigmatism disappears when using a point source, so it is the result of the final dimensions of the source).

Change the energy of the source from 35200 eV to 35700 eV and see the effect in focal position and intensity.

iii) Implement an elliptical lens. See the

Hints: You can use the workspaces i) `refractors_CRL_Snigirev_1996.ows`, ii) `refractors_transfocator.ows` and iii) `refractors_lens_elliptical.ows`.

Answer

i) See the script for calculate the refraction index and focal distances. Then, in the first branch we input externally the refraction index, and no absorption is used. The focal spot is 11 μm, and we see in the fistogram high tails. When using the internally calculated refraction index, that also includes reflectivity, we observe a 9 μm intensity profile with less tails. In third branch we recalculated the same CRL but using parabolic profile. We see limited improvements in this case, because the source is large and demagnification small. Playing with these parameters one can observe more drastic differences.

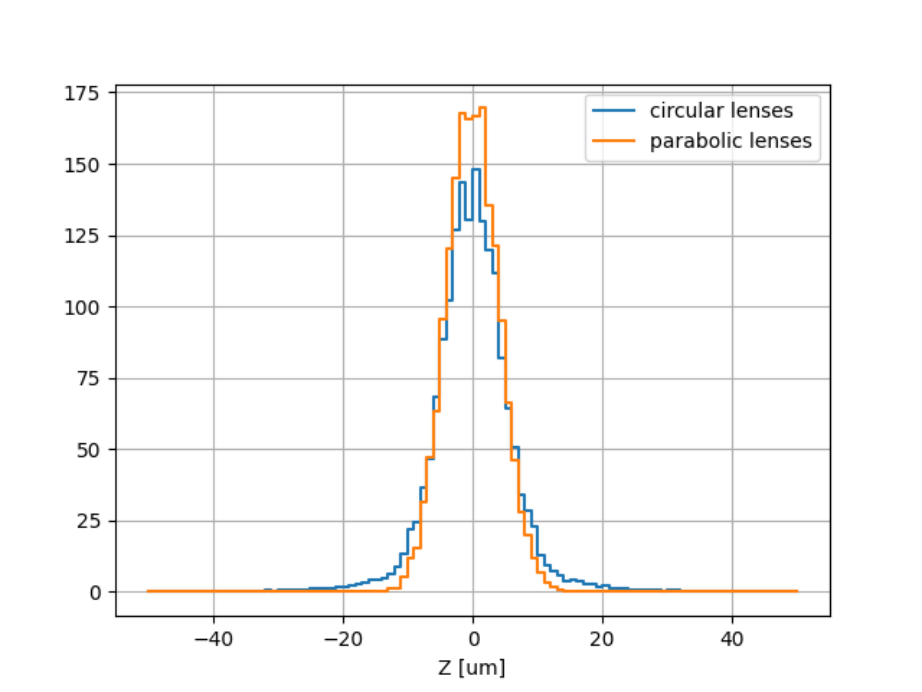


Figure 33: Intensity distribution in vertical direction at the image (focal) plane for a CRL with circular lenses and parabolic lenses.

ii) For E=35700 eV the focused beam has dimensions 32 μm × 6μm. The focal position is well centered, as seen with the “caustic" widget.

For E=35200 eV we find a focus of 33μm × 9μm (FWHM), best H focus at about -0.4 m.

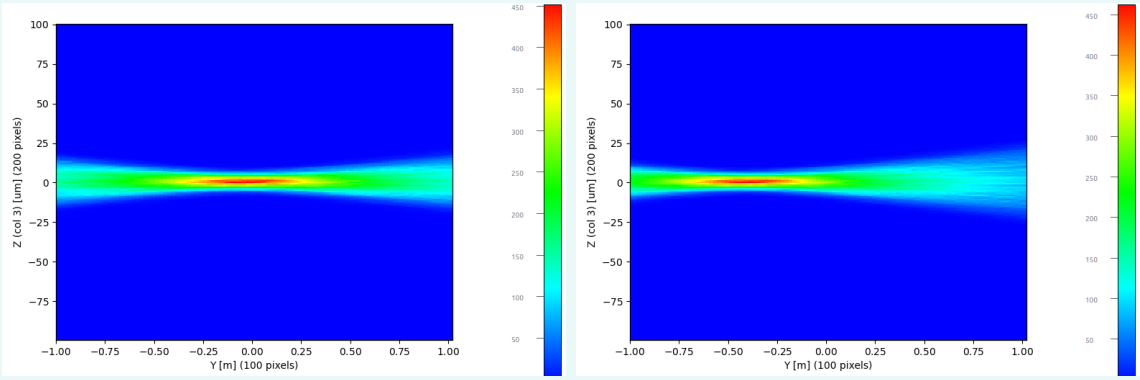


Figure 34: Caustic (vertical beam size vs propagation direction) at 35.7 keV (left) that is the energy used to tune the transfocator, and also at 35.2 keV (right) wher it can be observed a shift of the focal position at about 0.4 m upstream from the image plane. This illustrates the chromatic aberrations of the lenses.

iii) In the example workspace, we proceed as follow. We first use a collimated source, and in the different workflow branches we

- use the “Refractive Lens" widget to simulate a lens that will focus the collimated beam. Look at the conic coefficients that are automatically calculated
- implement these coefficients manually in the “Refractive Interface" widget. We split the lens in its two interfaces

- change from two parabolic interfaces to a plane plus another arabolic interface. The script calculates the parameters
- replace the parabolic lens for an elliptical lens

When comparing the results of the parabolic lens with the elliptical one, practically the same performance in terms of focal size. TO FIND ANOTHER SETUP WHERE DIFFERENCES CAN BE APPRECIATED

Then, it is used an almost point, and divergent source, and the parabolic and elliptical lenses are compared with the Cartesian Oval TO BE DONE!

14. Simulation of a complete beamline

14.1. Ray tracing simulation

You will:

- Combine several optical elements
- Obtain final results for a beamline in terms of flux, resolution and spot size

Questions:

Define the following elements in SHADOW that model an X-ray undulator line at the ESRF-EBS, operating at 5 keV, with the full photon transport path spanning 51 metres from source to sample.

- Source — Undulator U42 (period 42 mm) installed on a straight section of the EBS-ESRF storage ring. The on-axis brilliance peak is set to be near 5 keV. In SHADOW/OASYS this is represented either as an "Undulator Gaussian" (simple approximation for on-resonance emission) and also as an "Undulator Light Source" that samples rays according to the calculated undulator spectrum.
- Fron-end slits. All the fron-end is condensed in a single aperture of 2×2 mm^2 placed at 27.2 m from the source position.
- White beam mirrors — plane Ni pair of plane mirrors coated with nickel (Ni), set at a grazing incidence angle of 7 mrad. and deflecting the beam in the horizontal plane. The mirrors reflect efficiently the radiation at 5 keV, while acting as a low-pass filter — rejecting higher harmonics from the undulator spectrum.
- Monochromator — double-crystal Si111. The core energy-selecting element is a double-crystal monochromator with Si(111) Bragg reflection, deflecting in the vertical plane.
- Secondary slits, after monochromator. They define the beam that will be used at the Kirkpatrick–Baez (KB).
- Focusing system. The final focusing is achieved by a KB mirror system: two successive elliptical mirrors (one focusing vertically, one horizontally), each figured to an elliptical profile with the source at one focus and the sample at the other.
- The beam arrives at the sample as a tightly focused spot.

i) Calculate:

- Beam geometry at the sample position
- Energy resolution
- Transmitivity of the whole beamline, or number of photons at the sample position.

Hints: you may use the workspace file `beamline.ows`.

Answer

In the example workspace, the simulation is done in a first place using the "Undulator Gaussian". This is always a good starting point that gives a good approximation to the source emission on resonance.

Note also that the White mirrors deflect in the horizontal plane, therefore the *optical element orientation angles* are 90° and 180°, respectively. This result in a global rotation of the axis (after the two reflections, col1 or X is vertical and col3 or Z in horizontal). This is "corrected" using the "Empty element" after the white beam mirrors.

For the monochromators, the expected intrinsic resolution for the Si111 crystals is $\approx 10^{-4}$, therefore 0.5 eV at 5 keV. In consequence, the energy interval at the source has to be wide enough to cover the peak and tails. We selected ΔE_{src}=4 eV.

For the KB mirrors the calculation of the elliptical shape is made "internal/calculated", meaning that SHADOW computes it giving the position of the foci. Note that the mirrors are "elliptical cylinders", they are bent only in the tangential direction, i.e. 1D-focusing (each in a different direction), so this has to be defined in the mirror shape parameters. Note also that the second KB mirror (that focus in the horizontal) has *optical element orientation angle* 90° therefore a correction using "Empty element" is done upstream.

With this configuration (using "Undulator Gaussian") we obtain:

- Size: 39 × 32 nm^2
- Energy bandwidth ΔE=0.6 eV
- Flux at source: N_{src}=6.81×10^{14} photons/s/0.1%bw; Integrated Flux at sample N=2.15×10^{13} photons/s

The number of photons can be calculated Automatically, using the Info widget. or manually:

- At the source we have I_{src}=50000 (intensity=number of rays), ΔE_{src}= 4 eV, and intensity N_{src} = 6.82×10^{14} photons/s/0.1%bw=1.36×10^{14} photons/s/eV.
- The transmittivity of the beamline in one eV bandwidth is:

  $T = (I/\Delta E)/(I_{src}/\Delta E_{src})$= (1161/0.6)/(50000/4)=26.7%.
- the total number of phorons (in the full bandwidth) is $N = N_{src} \times T \times \Delta E$= 2.15×10^{13} photons/s.

With the configuration using "Undulator Light Source") we obtain:

- Size: 53 × 23 nm^2
- Energy bandwidth ΔE=0.52 eV

- Flux at source: N_{src}=3.25×10^{14} photons/s/0.1%bw; Integrated Flux at sample N=1.22×10^{13} photons/s

Note that there are some differences, that can be explained by the different values of flux at source. Indeed, the Undulator Gaussian used the analytical equation of flux at the central cone, which ignores effect of emittance. On the contrary, Undulator Light Source includes the emittance, which implies a red-shift of the peak, therefore the selected energy interval [4998,5002] does correspond to the resonance (therefore the emision is a central cone), but does not correspond to the peak of the spectrum.

Note also that the number of rays used is 15000, and the final intensity is 719. The number of rays should be increased to get better statistics (but needs more time and memory).

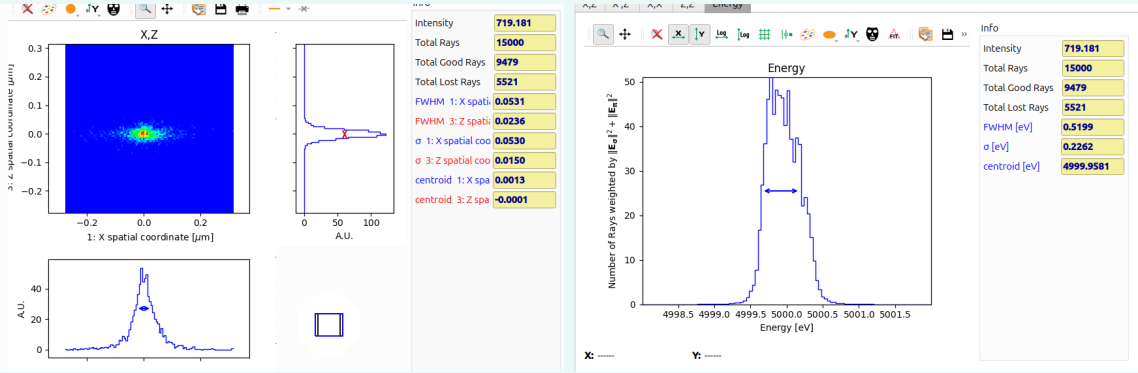


Figure 35: Ray tracing results in terms of intensity distribution in the (X,Z) plane at the image position (left), and energy resolution (right).

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